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## ABSTRACT

Accurately modeling polarization and van der Waals (vdW) interactions in atomistic systems typically requires high-level quantum-mechanical methods that are computationally expensive, hence limited in applicability. To address this challenge, efficient yet physically grounded models are needed—ones that not only enable accurate predictions but also provide insight into how noncovalent interactions scale in complex molecular and material systems. This review highlights the quantum Drude oscillator (QDO) model, a physically motivated and computationally efficient framework that captures the essential features of electronic response, including polarization and dispersion forces, across a wide range of chemical and material systems. We discuss how the QDO model quantitatively reproduces the polarization response of many-electron atoms and how key components of noncovalent interactions—exchange-repulsion, polarization, and dispersion—emerge naturally in QDO dimers. Furthermore, the model provides predictive scaling laws that elucidate trends in polarizability and dispersion across the periodic table and in molecular assemblies. By uniting interpretability, accuracy, and efficiency, the QDO model offers a versatile approach for modeling noncovalent interactions in systems ranging from isolated molecules to complex condensed phases and nanostructured materials.

## I. INTRODUCTION

Models based on (coupled) harmonic oscillators are widely employed in practical theoretical frameworks and for interpreting experimental data.[1] For example, molecular vibrations and phonons in solids are usually described by harmonic oscillators. In Feynman's path-integral formulation of quantum mechanics, the behavior of an arbitrary quantum system is *exactly* described by an infinite collection of coupled harmonic oscillators.[2] By construction, a single harmonic oscillator is capable of describing small oscillations of a system away from its equilibrium state.

In the context of intermolecular interactions,[3,4] polarization response and van der Waals (vdW) dispersion interactions in atomistic systems arise from fluctuations of electronic density around their equilibrium (mean-field) configuration. These oscillations are always present because static and/or dynamic electric fields permeate molecules and materials. Static fields that arise from surrounding nuclei/electrons affect atoms, while dynamic fields are specifically induced by electronic quantum zero-point fluctuations.[5] Both static and dynamic responses of atomistic systems are suitable for treatment via coupled harmonic oscillators. In principle, an infinite number of such oscillators would be required for an exact treatment of molecules and materials; hence, a practical question arises about how to parameterize the response of real atoms using harmonic oscillators. Would a *single* oscillator per atom suffice for a relatively accurate treatment? The study carried out in the last decade and summarized here suggests that a properly parameterized charged quantum harmonic oscillator, also called a *quantum Drude oscillator* (QDO), is able to accomplish such a difficult task.

Within the QDO model, each oscillator represents coarse-grained fluctuations in the charge density of weakly bound (valence) electrons in an atom. With a proper parameterization, the QDO model enables an efficient description of response properties and noncovalent interactions of atoms, small and large (bio)molecules, solids, nanostructures, and hybrid organic/inorganic interfaces.[6–17] This model provides an accurate description (within a few percentage points) of polarization and dispersion interactions.[6,13,14,18] It also accurately captures electron density redistribution induced by such interactions[19,20] and provides a robust tool to describe vdW interactions under the influence of external charges as well as spatial confinement.[21–23] Finally, the QDO model paves the way to obtain full binding energy curves of vdW-bonded dimers,[24] upon an appropriate generalization of the fermionic Pauli exchange to the coarse-grained QDO model.[16,17,25]

In this review, we focus on the theoretical underpinnings of the QDO model and explain the foundations for its, perhaps somewhat surprising, success in describing the polarization response and vdW interactions for an incredibly wide range of atomistic systems. The review is organized as follows. In Sec. II, we introduce the model, and then in Sec. III, we show how all four basic molecular interactions (electrostatics, polarization, dispersion, and exchange-repulsion) can be treated within the QDO model using perturbation theory (PT). Elegant atomic scaling laws and response properties, which can be obtained using this framework, are summarized in Sec. IV. Next, we discuss the non-perturbative approaches to solve the dipole- and full Coulomb-coupled QDO Hamiltonian and highlight the most important results in Sec. V. Section VI exemplifies the applications of the QDO model in atomistic simulations of molecular dimers and water. Finally, we discuss the possible extensions of the model to overcome existing limitations in Sec. VII.

## II. FRAMEWORK OF THE QDO MODEL

### A. Single QDO

Within the QDO model, the electronic response is described in a coarse-grained way by means of a *drudon*, a quasiparticle that effectively represents the valence electrons in an atom or molecule. The Hamiltonian of a single unperturbed QDO (labeled with A in Fig. 1) is

$$\hat{H}_0 = -\frac{\hbar^2}{2\mu}\nabla_{\mathbf{r}}^2 + \frac{1}{2}\mu\omega^2(\mathbf{r}-\mathbf{R}_A)^2, \tag{1}$$

where the two parameters $\omega$ and $\mu$ are the oscillator frequency and the mass of the drudon, respectively. In addition, the drudon acquires a negative electric charge $-q$ for electromagnetic interactions with other species or electric fields. The drudon is harmonically bound to its center of oscillation (pseudo-nucleus) having a positive charge $q$. It is important to note that the drudon and its own pseudo-nucleus do not interact via direct electromagnetic force. Instead, their interaction is fully described by the harmonic bond representing the time-dependent fluctuations of the electron density from its static equilibrium configuration. Therefore, every QDO is fully defined by its three parameters $\{q, \mu, \omega\}$.

### B. Coupled QDOs

For a system of two QDOs interacting via Coulomb forces, the Hamiltonian becomes

$$\hat{H}_{\text{Coul}} = \hat{H}_0^A + \hat{H}_0^B + \hat{V}_{\text{int}} = \hat{H}_0^A + \hat{H}_0^B + \frac{q_A q_B}{4\pi\varepsilon_0} \times\left(\frac{1}{R} + \frac{1}{|\mathbf{R}-\mathbf{r}_1+\mathbf{r}_2|} - \frac{1}{|\mathbf{R}-\mathbf{r}_1|} - \frac{1}{|\mathbf{R}+\mathbf{r}_2|}\right). \tag{2}$$

Here, the first two terms of $\hat{V}_{\text{int}}$ describe the interactions between two pseudo-nuclei and two drudons, respectively, whereas the last

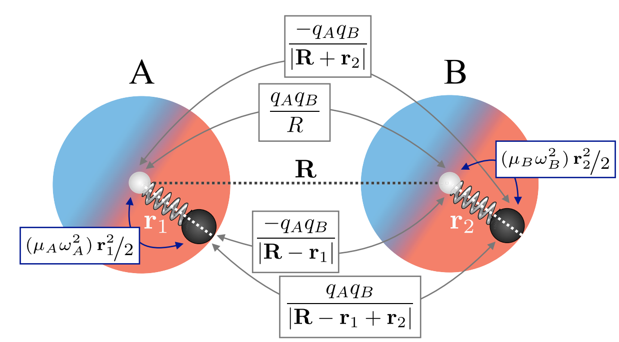


**FIG. 1.** Two QDOs separated by a distance *R*. The black and white spheres represent the two Drude particles and fixed nuclei, respectively. The Coulomb interactions between the two QDOs (gray arrows) and the harmonic intra-QDO potentials (blue arrows) are explicitly highlighted with their connection to Eqs. (1) and (2). Reproduced with permission from Vaccarelli *et al.*, Phys. Rev. Res. **3**, 033181 (2021). Copyright 2021 American Physical Society.

two terms correspond to the interactions between a drudon and the neighboring pseudo-nucleus, which is schematically illustrated in Fig. 1.

At a sufficiently long range, the Coulomb potential can be well approximated by the dipole coupling,

$$\hat{V}_{\rm dip} = \frac{q_A q_B}{4\pi\epsilon_0}\left(\frac{(\mathbf{r}_1\cdot\mathbf{r}_2)}{R^3} - \frac{3(\mathbf{r}_1\cdot\mathbf{R})(\mathbf{r}_2\cdot\mathbf{R})}{R^5}\right). \tag{3}$$

The Hamiltonian of dipole-coupled QDOs can be exactly diagonalized even under applied static electric fields,[26,27] which makes such a model widely applicable to large systems and allows relatively straightforward calculations of many response properties. The detailed discussion of intermolecular interactions in static electric fields within the QDO model can be found in Sec. III H.

The Hamiltonians of Eqs. (2) and (3) straightforwardly generalize to $N$ interacting QDOs. The methods to solve these—Quantum Monte Carlo (QMC),[28] Many-Body Dispersion (MBD),[7] and path-integral molecular dynamics (MD)[12,29]—are discussed in Sec. V. Moreover, one can go beyond the dipole approximation of the Coulomb interaction, which can be performed perturbatively for the case of atomic/molecular dimers[17,23] as well as non-perturbatively within the MBD method.[30–32]

## III. INTERMOLECULAR INTERACTIONS IN THE QDO MODEL FROM PERTURBATION THEORY

The leading-order intermolecular interactions can be broadly categorized into four types:[4] *electrostatics*, *exchange repulsion*, *polarization* (or *induction*), and *dispersion*. In the framework of a textbook perturbation theory,[4] electrostatics refers to the Coulomb interaction between static charge distributions, described by permanent multipoles at long range (e.g., dipole–dipole or dipole–quadrupole). The exchange repulsion force stems from the Pauli exclusion principle, which prevents same-spin electrons from occupying the same space. Induction describes the interaction between a static density and the density polarization it induces on a neighboring molecule. Finally, dispersion results from interactions between instantaneous multipoles generated by zero-point electron fluctuations, which induce corresponding multipoles in adjacent molecules. Notably, dispersion interactions are universally present, as they do not rely on permanent multipoles. The physical nature of these four interactions is visually depicted in Fig. 2.

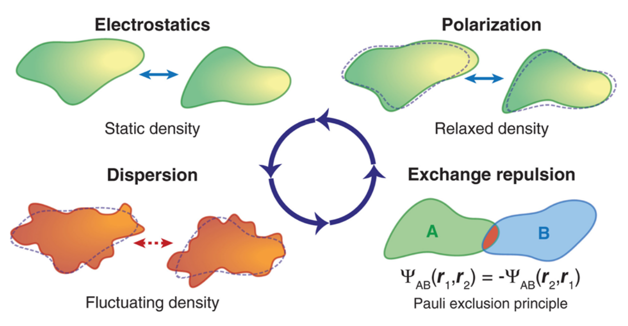


**FIG. 2.** Pictorial representation of noncovalent interactions (up to the second order in the Coulomb interaction). Note that "polarization" is also often termed "induction."[33,34] Reproduced with permission from J. Hoja, A. M. Reilly, and A. Tkatchenko, Wiley Interdiscip. Rev.: Comput. Mol. Sci. **7**, e1294 (2017). Copyright 2017 John Wiley and Sons, Ltd.[35]

In Secs. III A–III D, we summarize how these leading-order interactions can be treated within the QDO model. In Sec. III E, we discuss mixed dispersion–polarization interactions as an example of higher-order PT contribution. Section III F illustrates how the QDO model elucidates the origin of the so-called Feynman dispersion dipoles. In Sec. III G, we summarize the diagrammatic PT approach, enabling the general description of interactions beyond the second order PT. Finally, Sec. III H provides an account for QDO interactions in static electric fields.

### A. QDO model for electrostatics

The first-order perturbation energy between two QDOs interacting via Coulomb interaction in Eq. (2) reads[14]

$$E_{\rm C} = \frac{q^2}{4\pi\varepsilon_0}\left(\frac{1}{R} + \frac{\operatorname{erf}(R/2\sigma)}{R} - 2\frac{\operatorname{erf}(R/\sigma\sqrt{2})}{R}\right), \tag{4}$$

where

$$\sigma = \sqrt{\hbar/2\mu\omega}, \tag{5}$$

is the characteristic length of an oscillator. Physically, Eq. (4) describes the screening of Coulomb interactions due to the finite spatial extent of the QDO's Gaussian charge density. While real atomic charge distributions differ from the Gaussian profile assumed here, Eq. (4) nonetheless captures essential features of electrostatic interactions.[14] This functional form has recently found utility in physics-informed machine learning force fields (MLFFs) as a robust long-range energy correction.[36]

However, the direct application of Eq. (4) often leads to overestimated interaction strengths. This discrepancy arises because QDO parameters—particularly $\omega$ and $\mu$, and hence $\sigma$—are typically optimized to reproduce response properties (see Sec. IV C), rather than to match the actual spatial extent of atomic charge densities.[14] Consequently, the resulting effective size of the QDO can be too small, amplifying the Coulomb attraction between them. To mitigate this, the appropriate QDO reparameterization aimed at more accurate electrostatics is desirable and has been explored via data fitting in recent developments of MLFFs.[36]

Despite these limitations, existing response-based parameterizations remain quite effective in capturing the electrostatic *response* of atoms and molecules. For example, the QDO model reproduces the polarization potential around atoms remarkably well, as discussed in Sec. IV D. This underscores the strength of the QDO framework in modeling field-induced effects, even if further refinements are needed for interactions involving static charge distributions.

### B. QDO model for polarization

Multipole polarizabilities $\alpha_l$ are fundamental descriptors of the electronic response of atoms and molecules to external electric fields or the presence of interacting moieties. Within the QDO framework, these polarizabilities can be derived by considering the perturbing effect from a test charge placed at a large distance $D$ from the center of a QDO.[13] The resulting expressions for multipole polarizabilities are

$$\alpha_l = \left(\frac{q^2}{\mu\omega^2}\right)\left[\frac{(2l-1)!!}{l}\right]\left(\frac{\hbar}{2\mu\omega}\right)^{l-1}. \tag{6}$$

These results connect the QDO parameters directly to physically measurable quantities. Specifically, the dipole, quadrupole, and octupole polarizabilities can be recursively expressed as[13]

$$\alpha_1 = \frac{q^2}{\mu\omega^2}, \quad \alpha_2 = \frac{3}{4}\left(\frac{\hbar}{\mu\omega}\right)\alpha_1, \quad \alpha_3 = \frac{5}{4}\left(\frac{\hbar}{\mu\omega}\right)^2\alpha_1. \tag{7}$$

As evident from Eq. (6), higher-order polarizabilities depend explicitly on $\hbar$, highlighting their intrinsically quantum-mechanical origin. In the classical limit $\hbar \to 0$, all $\alpha_l$ with $l > 1$ vanish, leaving only the dipole polarizability $\alpha_1$.[13,37]

From these expressions, one can define the dimensionless polarizability invariant,[13]

$$\gamma_{\text{pol}} = \sqrt{\frac{20}{9}}\frac{\alpha_2}{\sqrt{\alpha_1\alpha_3}} = 1. \tag{8}$$

This invariant quantifies the internal consistency of the QDO model across different orders of polarizability. When compared to experimental values of $\gamma_{\text{pol}}$ for atoms and small molecules (see Fig. 3), the model shows striking agreement, especially for noble gases and hydrogen. Deviations are more pronounced for alkali atoms, where the error can reach up to 20%, likely due to their more diffuse electron densities and stronger many-electron effects.

These results underscore the power and simplicity of the QDO model in capturing essential features of the electronic response. In Sec. III H, we extend this discussion by exploring how QDOs respond to external polarizing fields and how this underpins their ability to model polarization-driven intermolecular interactions.

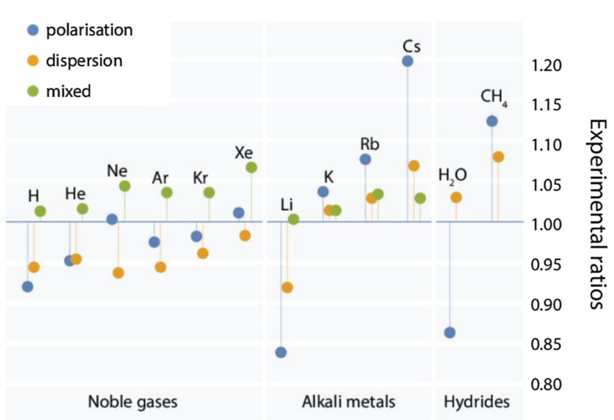


**FIG. 3.** QDO invariants for polarization, $\gamma_{\text{pol}}$ of Eq. (8), and dispersion, $\gamma_{\text{disp}}$ of Eq. (11), as well as the mixed invariant, $\gamma_{\text{3disp}}$ of Eq. (12). Reproduced with permission from Cipcigan *et al.*, J. Comput. Phys. **326**, 222 (2016). Copyright 2016 Elsevier Inc.

## C. QDO model for vdW dispersion

Beyond its success in describing polarizabilities, the QDO model enables closed-form calculations of vdW dispersion coefficients, first demonstrated by London in his seminal study on vdW dispersion forces.[38] For homonuclear dimers, the leading-order dispersion coefficients are given by[13]

$$C_6 = \frac{3}{4}\alpha_1^2\hbar\omega, \quad C_8 = \frac{5\hbar}{\mu\omega}C_6, \quad C_{10} = \frac{245\hbar^2}{8(\mu\omega)^2}C_6. \tag{9}$$

These expressions can also be recast in terms of QDO-derived multipole polarizabilities,

$$C_8 = 5\alpha_1\alpha_2\hbar\omega, \quad C_{10} = \left(\frac{21}{2}\alpha_1\alpha_3 + \frac{35}{2}\alpha_2\alpha_2\right)\hbar\omega. \tag{10}$$

These scaling laws reveal the natural hierarchy of dispersion interactions and have been shown to be applicable to real atoms and molecules with remarkable accuracy.[6,13,18]

Analogous to the invariant introduced for polarizabilities, the QDO model yields a dispersion invariant defined as[13]

$$\gamma_{\text{disp}} = \sqrt{\frac{49}{40}}\frac{C_8}{\sqrt{C_6C_{10}}} = 1. \tag{11}$$

This expression reflects the internal consistency of the QDO model's predictions for dispersion coefficients across different orders.

A further invariant—bridging polarization and dispersion—can also be introduced as[13]

$$\gamma_{\text{3disp}} = \frac{\alpha_1 C_6}{4C_9} = 1, \tag{12}$$

where $C_9 = \alpha_1 C_6/4$ is the three-body dispersion coefficient within the QDO model, commonly known as the Axilrod–Teller–Muto term.[13] Figure 3 shows experimental data for real atoms and molecules matching quite well the QDO invariants from Eqs. (11) and (12).

The QDO-based dispersion correction schemes are among the most successful and physically grounded approaches for recovering the missing long-range dispersion interactions in practical implementations of density functional theory (DFT).[6,40,41] At the heart of these models lies the renowned London formula,[38] $C_6 = \frac{3}{4}\hbar\omega\alpha_1^2$, which relates the leading-order van der Waals dispersion coefficient to a system's characteristic frequency and dipole polarizability. Despite its conceptual simplicity, this formula remains central to modern dispersion models. A notable example is the Tkatchenko–Scheffler (TS) method,[42] which combines the London formula with free-atom reference data and ground-state electron densities, yielding molecular $C_6$ dispersion coefficients with a mean absolute error (MAE) of just 5.5% when compared to reference vdW coefficients derived from experimental dipole-oscillator measurements. Building upon this, the dipole-coupled QDO framework further led to the development of the MBD method,[7] which extends pairwise dispersion to collective many-body interactions. This methodology is further discussed in Sec. V D.

Overall, the QDO model provides not only conceptual clarity but also practical tools for accurately describing vdW forces, making it a cornerstone in the development of physics-based dispersion correction schemes.

### D. Exchange-repulsion in the QDO model

The original QDO model[10,11,13] did not account for exchange-repulsion effects due to two main limitations. First, the coarse-grained wave function of a single Drude particle collectively represents all valence electrons in an atom, preventing a straightforward treatment of Pauli exchange between individual electron pairs.[43] Second, Drude particles are fundamentally distinguishable,[13] each tied to a specific nucleus and hence experiencing a distinct physical potential.[28] Here, we show how exchange-repulsion between identical QDOs can be effectively captured via the Heitler–London approach,[44] yielding a coarse-grained model for Pauli repulsion between real atoms.

Consider a homonuclear dimer of closed-shell atoms $A$ and $B$, represented by two identical QDOs. Let atom $A$ be located at the origin, while atom $B$ is displaced by a vector $\mathbf{R}$. Their ground-state QDO wave functions are

$$\begin{aligned}\psi_A(\mathbf{r}_1) &= \left(\frac{\mu\omega}{\pi\hbar}\right)^{3/4} e^{-\frac{\mu\omega}{2\hbar}r_1^2},\\ \psi_B(\mathbf{r}_2) &= \left(\frac{\mu\omega}{\pi\hbar}\right)^{3/4} e^{-\frac{\mu\omega}{2\hbar}(\mathbf{r}_2-\mathbf{R})^2},\end{aligned} \tag{13}$$

where the drudon coordinates $\mathbf{r}_1$ and $\mathbf{r}_2$ are measured from the common origin at atom $A$. A simple product wave function, $\Psi_{\rm dis}(\mathbf{r}_1,\mathbf{r}_2) = \psi_A(\mathbf{r}_1)\psi_B(\mathbf{r}_2)$, implies distinguishable particles. However, for identical (indistinguishable) closed-shell atoms, a symmetrized product must be used as a *collective* QDO wave function,

$$\Psi_{\rm ind}(\mathbf{r}_1,\mathbf{r}_2) = \frac{\psi_A(\mathbf{r}_1)\psi_B(\mathbf{r}_2) + \psi_A(\mathbf{r}_2)\psi_B(\mathbf{r}_1)}{\sqrt{2(1+S)}}, \tag{14}$$

where $S$ is the overlap integral,

$$S = \int\int d\mathbf{r}_1 d\mathbf{r}_2\ \psi_A^*(\mathbf{r}_1)\psi_B^*(\mathbf{r}_2)\psi_A(\mathbf{r}_2)\psi_B(\mathbf{r}_1) = e^{-\frac{\mu\omega}{2\hbar}R^2}. \tag{15}$$

Unlike Ref. 16, we use the normalized indistinguishable-particle wave function, $\langle\Psi_{\rm ind}|\Psi_{\rm ind}\rangle = 1$, in alignment with the original Heitler–London approach.[44]

Since, with QDOs, we aim to model the *collective* exchange-repulsion, the effective QDO interaction potential employed to capture this (long-range) exchange-repulsion is best represented via a multipole expansion, offering a coarse-grained, collective description of electrons.[16,17] Due to their coarse-grained nature, QDOs cannot resolve individual electron-pair exchanges. We thus adopt the dipole approximation[16] using coordinates measured from a common origin, as per Eq. (13). The extensions of higher-order multipole contributions are straightforward and discussed in Ref. 17. The derivation below complements and refines that of Ref. 16.

The exchange-repulsion energy, following Heitler and London,[44] is the difference between the first-order PT corrections evaluated on $\Psi_{\rm ind}$ and $\Psi_{\rm dis}$,

$$\begin{aligned}E_{\rm ex} &= \langle\Psi_{\rm ind}|\hat{V}_{\rm dip}|\Psi_{\rm ind}\rangle - \langle\Psi_{\rm dis}|\hat{V}_{\rm dip}|\Psi_{\rm dis}\rangle\\ &= \frac{C_{\rm ind}+J_{\rm ex}}{1+S} - C, \quad \text{with } C_{\rm ind} = \frac{C+C'}{2}.\end{aligned} \tag{16}$$

Here, $J_{\rm ex}$ and $C$ are the standard exchange and Coulomb integrals, respectively,

$$J_{\rm ex} = \langle\psi_A(\mathbf{r}_1)\psi_B(\mathbf{r}_2)|\hat{V}_{\rm dip}|\psi_A(\mathbf{r}_2)\psi_B(\mathbf{r}_1)\rangle, \tag{17}$$

$$C = \langle\psi_A(\mathbf{r}_1)\psi_B(\mathbf{r}_2)|\hat{V}_{\rm dip}|\psi_A(\mathbf{r}_1)\psi_B(\mathbf{r}_2)\rangle, \tag{18}$$

with $\hat{V}_{\rm dip}$ as the dipolar interaction potential,

$$\hat{V}_{\rm dip} = \frac{q_A q_B}{4\pi\epsilon_0}\left[\frac{(\mathbf{r}_1\cdot\tilde{\mathbf{r}}_2)}{R^3} - \frac{3(\mathbf{r}_1\cdot\mathbf{R})(\tilde{\mathbf{r}}_2\cdot\mathbf{R})}{R^5}\right], \tag{19}$$

where we introduced $\tilde{\mathbf{r}}_2 = \mathbf{r}_2 - \mathbf{R}$ for compactness. The Coulomb integral with exchanged particles is

$$C' = \langle\psi_A(\mathbf{r}_2)\psi_B(\mathbf{r}_1)|\hat{V}'_{\rm dip}|\psi_A(\mathbf{r}_2)\psi_B(\mathbf{r}_1)\rangle, \tag{20}$$

with the symmetrized operator

$$\hat{V}'_{\rm dip} = \frac{q_A q_B}{4\pi\epsilon_0}\left[\frac{(\tilde{\mathbf{r}}_1\cdot\mathbf{r}_2)}{R^3} - \frac{3(\tilde{\mathbf{r}}_1\cdot\mathbf{R})(\mathbf{r}_2\cdot\mathbf{R})}{R^5}\right], \tag{21}$$

where we defined $\tilde{\mathbf{r}}_1 = \mathbf{r}_1 - \mathbf{R}$. This preserves the Coulomb integral physically meaningful even when drudons are displaced from their original pseudonuclei, as in Eq. (20). Notably, $J_{\rm ex}$ is invariant to the choice of coupling from Eqs. (19) or (21).

Explicit integration yields[16]

$$J_{\rm ex} = \frac{q^2 S}{(4\pi\epsilon_0)2R} = \frac{q^2 e^{-\frac{\mu\omega}{2\hbar}R^2}}{(4\pi\epsilon_0)2R} \quad \text{and} \quad C = C' = 0. \tag{22}$$

Substituting Eq. (22) into Eq. (16) yields

$$E_{\rm ex} = \frac{J_{\rm ex}}{1+S} \approx J_{\rm ex}, \tag{23}$$

as the overlap integral $S = e^{-\frac{\mu\omega}{2\hbar}R^2} \ll 1$ for vdW-bonded dimers near equilibrium.[16]

Therefore, $J_{\rm ex}$ of Eq. (22) provides a reliable approximation for exchange-repulsion energy in the QDO framework and offers an alternative to existing analytical models based on overlapping atomic densities[45,46] or orbitals.[47] This result was recently used to construct universal interatomic vdW potentials based on the QDO model,[24] as discussed in Sec. VI D. We note that, while derived for homonuclear dimers, $J_{\rm ex}$ of Eq. (22) can be transferred to heteronuclear systems using combination rules, as demonstrated in Ref. 24.

### E. Dispersion-polarization within the QDO model

As part of the electronic energy, vdW dispersion also affects the equilibrium electronic charge distribution. This effect, which we will dub *dispersion-polarization* in analogy to the corresponding term appearing in third-order symmetry-adapted perturbation theory (SAPT),[33,48] naturally also contributes to inter- and intramolecular (interaction) energies. While the role of dispersion–polarization has been found to be largely negligible for the structure and interaction energy of small organic molecules and dimers,[49–52] its contribution does become increasingly relevant for more polarizable systems such as alkali-metal complexes, bare metal surfaces, or extended organic molecules on metallic surfaces.[51,53]

Accurately describing this effect generally requires a self-consistent treatment of long-range (dynamic) electron correlation within the electronic structure method or going beyond the usual second-order truncation in SAPT—both of which are computationally expensive. In contrast, the QDO model provides a more efficient alternative. While it is well-established that QDOs can capture both dispersion and polarization independently (*vide supra*), it has been shown that the density difference of interacting and non-interacting QDOs can also capture the charge polarization due to dispersion interactions.[19,20] A recent study highlighted how the effect of this interplay (interaction) on energies—dispersion–polarization—can be accessed as well within the QDO framework via the perturbative description of the Coulomb interaction between quantum fluctuations obtained from dipole-coupled oscillators.[22,23]

Therefore, the QDO model offers the efficient-yet-reliable approach needed to perform a more rigorous investigation of the interplay between dispersion and polarization in extended and complex systems. One promising implementation of this idea is discussed in Sec. V E, where we present an approach grounded in the QDO framework that captures this interplay with both efficiency and physical rigor.

### F. Elucidating the origin of Feynman dispersion dipoles

The interplay between dispersion and polarization discussed earlier naturally brings us to the concept of the so-called Feynman dispersion dipoles. While dispersion interactions are often described[4,6,40] as arising from instantaneous interactions between fluctuating dipoles associated with atoms and molecules, Feynman[54] offered a different interpretation. He argued that dispersion induces permanent local, atom-centered dipole moments of order $1/R^7$, and that it is "*the attraction of each nucleus for the distorted charge distribution of its own electrons that gives the attractive $1/R^7$ force*."[54]

These seemingly different pictures of dispersion interactions were reconciled by Hunt,[55] who proved analytically that the long-range vdW force results from the Coulomb attraction of each nucleus to the change in the electronic charge density induced by dispersion. An earlier study by Hunt and Bohr[56] provided a full explanation of these dispersion dipoles, accounting for hyperpolarization and the modulation of spontaneous charge fluctuations by external fields. The resulting dipoles depend on the dipole–dipole–quadrupole hyperpolarizability, $B$.[56–58]

Subsequently, Odbadrakh and Jordan[59] showed that for two interacting QDOs, the Feynman dipoles can be obtained from the expectation value of the dipole operator using the second-order wave function, as well as using the response function approach. To obtain the dispersion dipoles, it is necessary to include both the dipole and quadrupole terms in the interaction between the two QDOs. Both approaches give $(9\omega/4R^7)\alpha_A B_B^{zzzz}$ (in atomic units) for the dipole induced on oscillator $B$ by a dipole fluctuation of oscillator $A$. In this expression, $\omega$, $\alpha_A$, and $B_B$ are the frequency, the dipole polarizability, and the dipole–dipole–quadrupole hyperpolarizability of a QDO, respectively.

It is important to note that the first-order wave function, which is used in the standard perturbative definition of dispersion,[33,34] does not capture the Feynman dipoles, and that the second-order terms in the wave function are essential for describing these distortions. The spontaneous, purely quantum-mechanical fluctuations in the dipoles on each of the interacting atoms or molecules produce a field that polarizes every interaction partner via induction. It is this interaction of the quantum fluctuations with the induced dipoles that gives rise to vdW dispersion forces, not the interactions of the spontaneously fluctuating dipoles themselves.

### G. Diagrammatic perturbation theory for the QDO model

The intermolecular interactions discussed in Secs. III A–III D emerge naturally within the first and second orders of perturbation theory (PT). However, the QDO model supports a much richer hierarchy of long-range dispersion and polarization effects (exemplified in Secs. III E and III F) that arise at higher PT orders and involve increasingly complex particle correlations—extending beyond simple two-body terms to intricate three-body and many-body interactions.

These contributions can be systematically organized and represented using diagrammatic techniques, similar to Feynman diagrams in perturbative quantum field theory. Such an approach for QDOs was pioneered by Jones *et al.*,[13] who introduced a compact, visually intuitive representation, elucidating the hierarchy of many-body responses and feedback loops emerging naturally within the coupled QDO model. This framework is inspired by the Jastrow diagrammatic expansion originally devised to incorporate correlations in many-body systems, thereby systematically improving upon mean-field approximations.[60]

The diagrammatic framework developed in Ref. 13 and illustrated in Fig. 4 is particularly useful for four reasons: (1) it identifies

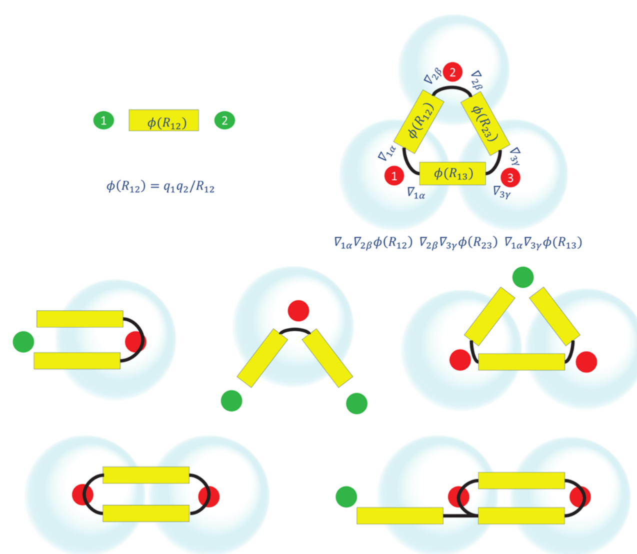


**FIG. 4.** Diagrams show selected terms among the hierarchy of QDO interactions and responses. **Top row:** The simple Coulomb potential between two fixed charges (left) and three-body dipole dispersion (right). **Middle row:** Dipole polarization by a fixed charge (left), dipole polarization by two charges at a single center (middle), and many-body polarization (right). **Bottom row:** Two-body dispersion loop (left) and mixed polarization–dispersion (right). Please see the text for a more detailed explanation. Reproduced with permission from Jones *et al.*, Phys. Rev. B **87**, 144103 (2013). Copyright 2013 American Physical Society.

those responses that are absent by symmetry, (2) it organizes physical responses by their order in PT, (3) it introduces dispersion as loops, drawing a formal analogy with the "bubble" diagrams from particle physics, where such loops contribute to the self-energy of a quantum field—in this case, however, the quantum field is composed of discrete QDOs localized at molecular sites, rather than being represented as a continuous field on a lattice, and (4) it captures complex mixed interactions, such as field-induced modulation of dispersion interactions. These diagrams, therefore, illustrate the responses of the model systematically without having to write or compute every term explicitly.

Figure 4 shows illustrative diagrams representing QDOs (blue clouds) with their tethering centers (red dots) and permanent charges (green). Polarization is visually represented as a displacement of the blue cloud from its red tethering center. Yellow bars indicate primitive electronic multipole interactions, and the number of thin black lines emanating from one end of a yellow bar denotes the multipole order (0 = monopole, 1 = dipole, 2 = quadrupole, etc.).

The two cases shown in Fig. 4 (top right and bottom left) are low-order examples of diagrams where all ends of the (yellow) boxes are connected. These loops correspond to pure dispersion energies that arise even in the absence of external fields at second and higher orders of PT. The bottom right diagram in Fig. 4 shows the polarizing effect of external charge on dispersion interaction between two QDOs (*field-induced dispersion*). This effect was studied quantitatively by Kleshchonok and Tkatchenko,[21] who found that the field-induced dispersion contribution can account for up to 35% of the total intermolecular binding energy in specific environments.

Importantly, the diagrammatic formalism is systematically extensible; it offers a well-defined path toward evaluating arbitrary high-order mixed responses, providing a theoretical foundation for exploring novel quantum many-body effects within the QDO framework.

### H. Intermolecular interactions in static electric fields within the QDO model

The complete set of eigenstates of a QDO in a uniform electric field enables expanding perturbed states of the coupled QDO–field system under the influence of linear perturbations, e.g., describing interactions with nearby QDOs as well as with macroscopic bodies and boundary conditions. In turn, such an expansion enables studying retarded and non-retarded intermolecular interactions under the influence of a variety of boundary conditions and/or external fields using the perturbative approaches within quantum mechanics and quantum electrodynamics (QED) theories.[61–66]

In quantum mechanics, the Hamiltonian of two interacting QDOs in dipole approximation under a static electric field $\boldsymbol{\mathcal{E}}(\mathbf{r})$ is expressed as

$$\hat{H} = \sum_{A=1,2} \hat{H}_{\mathcal{E}_A} + \hat{V}_{\text{int}}, \tag{24}$$

where $\boldsymbol{\mathcal{E}}_A$ is the external electric field at the center of the $A$th QDO and $\hat{V}_{\text{int}}$ is the dipole interaction from Eq. (3). $\hat{H}_{\mathcal{E}_A}$ is the Hamiltonian of a single QDO in field

$$H_{\mathcal{E}_A} = -\frac{\hbar^2}{2\mu_A}\nabla_A^2 + \frac{1}{2}\mu_A\omega_A^2\mathbf{r}_A^2 - q_A\mathbf{r}_A\cdot\boldsymbol{\mathcal{E}}_A, \tag{25}$$

which can be solved exactly. The eigenstates are the shifted Hermite functions $\phi_{\mathbf{n}}$

$$\psi_{\mathbf{n}}(\mathbf{r}) = \phi_{\mathbf{n}}(\mathbf{r} - q\boldsymbol{\mathcal{E}}/m\omega^2), \quad \mathbf{n} = (n_x, n_y, n_z), \tag{26}$$

with the corresponding energy levels

$$E_{\mathbf{n}} = \hbar\omega(n_x + n_y + n_z + 3/2) - q\,\mathcal{E}^2/2m\omega^2, \tag{27}$$

which is the spectrum of a quantum harmonic oscillator with the Stark shift.

Using the non-interacting QDO-field states, the interaction energy due to $V_{\text{int}}$ is straightforwardly computed via perturbation theory. Since the applied external fields break the spherical symmetry, the first-order correction no longer vanishes, yielding the electrostatic interaction energy $\Delta E_{\text{es}}^{(\text{QM})}$ between two field-induced dipole moments that scales with inter-QDO distance as $R^{-3}$. Second-order perturbation produces a field-induced polarization interaction $\Delta E_{\text{pol}}^{(\text{QM})}$ in addition to the well-known dispersion interaction $\Delta E_{\text{disp}}^{(\text{QM})}$, both scaling as $R^{-6}$.[26,27]

In the dipole approximation, the Hamiltonian of Eq. (24) can be exactly diagonalized to obtain a system of two decoupled QDOs in the normal-mode space (generalizable to $N$ interacting QDOs).[27] Hence, the interaction energy of the coupled QDO-field systems is calculated as the difference between the ground-state energy of the coupled normal modes and the sum of ground-state energies of the two non-interacting QDO-field systems given by Eq. (27).

The QDO model also facilitates studies of intermolecular interactions in static fields within QED, including retardation effects or coupling to fluctuating vacuum fields (e.g., cavities). In the multipolar formalism of QED, atoms interact with the vacuum field via their electric and/or magnetic moments and emit/absorb virtual photons. The exchange of such photons between atoms is understood as interatomic interactions.[67,68] External electric fields, inducing electric multipole moments, enable additional atomic degrees of freedom for interactions with the vacuum field, hence altering interatomic interactions. Since the QDO model provides exact solutions for an atom in a static field within the dipole approximation, the problem reduces to the coupling of QDOs with the vacuum field through a fluctuating dipole moment and a dipole moment induced by the static field.[27]

The Hamiltonian of two interacting QDOs under a static field within multipolar QED reads

$$\hat{H} = \sum_{A=1,2} \hat{H}_{\mathcal{E}_A} + \hat{H}_{\text{rad}} - \sum_{A=1,2} q_A\mathbf{r}_A\cdot\boldsymbol{\mathcal{E}}_\perp(\mathbf{R}_A), \tag{28}$$

with $\hat{H}_{\mathcal{E}_A}$ as in Eq. (25), $\hat{H}_{\text{rad}}$ the vacuum field Hamiltonian, and the last term describing the interaction of QDO dipoles with the transverse vacuum electric field $\boldsymbol{\mathcal{E}}_\perp$,[67,68]

$$\boldsymbol{\mathcal{E}}_\perp(\mathbf{r}) = i\sum_{\mathbf{k},\lambda}\sqrt{\frac{\hbar ck}{2V\varepsilon_0}}\Big[\mathbf{e}_{\mathbf{k}\lambda}\hat{a}_{\mathbf{k}\lambda}e^{i\mathbf{k}\cdot\mathbf{r}} - \bar{\mathbf{e}}_{\mathbf{k}\lambda}\hat{a}^\dagger_{\mathbf{k}\lambda}e^{-i\mathbf{k}\cdot\mathbf{r}}\Big]. \tag{29}$$

Here, $\hat{a}_{\mathbf{k}\lambda}$ ($\hat{a}^\dagger_{\mathbf{k}\lambda}$) is the annihilation (creation) operator of a photon of the mode $(\mathbf{k},\lambda)$ with the unit polarization vector $\mathbf{e}_{\mathbf{k}\lambda}$ inside the quantization volume $V$ (eventually $V \to \infty$).

The energy shift of the QDOs–radiation system due to their interaction is further evaluated via perturbation theory, with the unperturbed states given by the products of Fock states of the radiation field and the QDO-field states of Eq. (26). Since the transverse field is linear in the bosonic operators, only even orders of perturbation survive. The second-order correction is non-vanishing only when field-induced dipoles of QDOs interact with the transverse field and exchange one virtual photon, which results in the field-induced electrostatic interaction energy $\Delta E_{\mathrm{es}}^{(\mathrm{QED})}$.[27]

In the fourth order, interactions of both fluctuating and field-induced dipole moments with $\boldsymbol{\mathcal{E}}_{\perp}$ can contribute to the energy shift when QDOs exchange two virtual photons. When both QDOs interact with the vacuum field via their fluctuating dipole moments, the resulting interaction in the non-retarded regime is the well-known dispersion energy $\Delta E_{\mathrm{disp}}^{(\mathrm{QED})} \propto R^{-6}$ ($\propto R^{-7}$ in the retarded case), while the coupling of $\boldsymbol{\mathcal{E}}_{\perp}$ with a field-induced dipole moment at one center and a fluctuating dipole moment at the other center leads to a polarization interaction energy $\Delta E_{\mathrm{pol}}^{(\mathrm{QED})} \propto R^{-6}$ (valid in both retarded and non-retarded regimes). In free space and the non-retarded limit, the sum of these energy shifts recovers the sum of their QM counterparts $\Delta E_{\mathrm{es}}^{(\mathrm{QM})}$, $\Delta E_{\mathrm{pol}}^{(\mathrm{QM})}$, and $\Delta E_{\mathrm{disp}}^{(\mathrm{QM})}$.[27]

Before ending this subsection, it is worth mentioning that in a homogeneous electric field, a single QDO's ground-state density undergoes a rigid displacement, leading to vanishing $\beta$ and $\gamma$ hyperpolarizabilities.[13] Hence, some interactions stemming from ($\beta$ and $\gamma$) hyperpolarization of molecules in electric fields[56,69–75] also vanish for coupled QDOs. However, terms originating from static field-induced $\beta$ hyperpolarizability[76,77] decay more rapidly with distance, while those linked to dispersion-induced polarizability changes[78–80] are typically negligible compared to the long-range linear response terms, such as $\Delta E_{\mathrm{es}}$, $\Delta E_{\mathrm{pol}}$, and $\Delta E_{\mathrm{disp}}$. The role and relative importance of such nonlinear response contributions—especially from the $\gamma$ hyperpolarizability—are analyzed in detail in Sec. VII A.

## IV. QDO-DERIVED SCALING LAWS AND INSIGHTS FOR ATOMS AND MOLECULES

As demonstrated in Sec. III, the QDO model offers a unified analytical framework to describe various intermolecular interactions. Over the past decade, this has led to the discovery of non-trivial scaling laws for atomic vdW radii[16,17] and multipole polarizabilities,[16,17,81,82] summarized in Secs. IV A and IV B. Accurate predictions within the QDO framework require careful parameterization (Sec. IV C), exemplified here by modeling atomic response to electric fields[18] (Sec. IV D). Last but not least, Sec. IV E explores the connection between the QDO and exchange-hole dipole moment (XDM) models.

### A. Polarizability vs vdW radius

The QDO model's equal-footing treatment of repulsive and attractive forces enabled an analytical quantum-mechanical model for noble-gas dimers.[16] Inspired by Tang and Toennies,[43,89] equilibrium in these dimers is described as a balance between exchange repulsion and dispersion attraction. Within this framework, the vdW radius scales non-trivially with the dipole polarizability,[16]

$$R_{\mathrm{vdW}} = 2.54\ \alpha_1^{1/7} \quad \text{a.u.}. \tag{30}$$

The proportionality constant in Eq. (30), originally fitted to noble gases [Fig. 5(a)], was later derived heuristically in Ref. 85, yielding

$$\alpha_1 = (4\pi\epsilon_0/a_0^4)\ \alpha_{\mathrm{FSC}}^{4/3}\ R_{\mathrm{vdW}}^7, \tag{31}$$

where $\alpha_{\mathrm{FSC}}$ is the fine-structure constant. Figure 5(b) confirms the validity of Eq. (31) across the periodic table, establishing a universal definition of vdW radius based on dipole polarizability.

This success is notable, given that higher-order multipole terms contribute up to 30% of the equilibrium vdW energy.[6] A likely



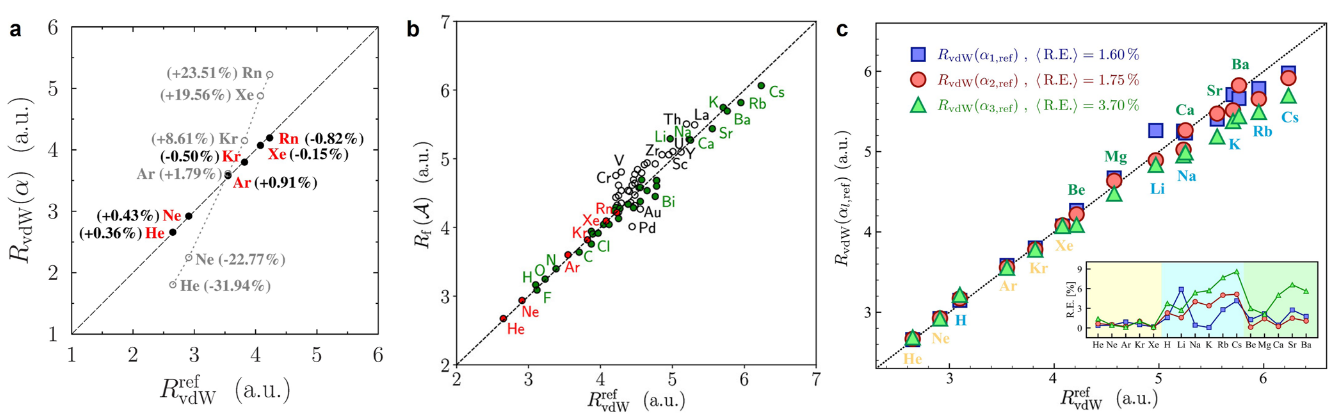


**FIG. 5.** (a) vdW radii obtained for noble gases by Eq. (30) in comparison to their Ref. 83 values (black full dots). The fit of the classical scaling law to the reference data, $R_{\mathrm{vdW}}(\alpha) = 1.62\alpha^{1/3}$, is shown by gray circles. The relative errors are given in parentheses. Adapted with permission from Fedorov *et al.*, Phys. Rev. Lett. **121**, 183401 (2018). Copyright 2018 American Physical Society. (b) vdW radii obtained for 72 atoms in the periodic table by Eq. (31) in comparison to their literature values from Refs. 83 and 84. Reproduced with permission from A. Tkatchenko, D. V. Fedorov, and M. Gori, J. Phys. Chem. Lett. **12**, 9488 (2021). Copyright 2021 American Chemical Society. (c) The vdW radius as a function of reference multipole polarizabilities[13,86–88] obtained according to Eq. (32) and shown vs the literature values $R_{\mathrm{vdW}}^{\mathrm{ref}}$.[42,83,84] Adapted from Vaccarelli *et al.*, Phys. Rev. Res. **3**, 033181 (2021).Copyright 2021 American Physical Society.

explanation is the partial cancellation of higher-order terms, as supported by the generalized relationship[16,17]

$$R_{\mathrm{vdW}}(\alpha_l) = A_l\, \alpha_l^{2/7(l+1)}, \quad l = 1, 2, 3, \tag{32}$$

with $A_1 = 2.54$, $A_2 = 2.45$, and $A_3 = 2.27$ in atomic units. Applied to experimental $\alpha_2$ and $\alpha_3$ for He–Xe,[13] Eq. (32) provides the relative errors below 1.4% [Fig. 5(c)], comparable to Eq. (30).[16]

The quantum-mechanical relations in Eqs. (30) and (31) contrast with the classical relation [see Fig. 5(a)] derived from the conducting spherical shell[90] and hard sphere[91] models,

$$R_{\mathrm{vdW}}^{\mathrm{class}} = (\alpha_1/4\pi\epsilon_0)^{1/3}. \tag{33}$$

This classical relation was widely used until the quantum-mechanical relation was established in Refs. 16 and 85.

For instance, effective vdW radii $R_{\mathrm{vdW}}^{\mathrm{eff}}$, used in damping functions for DFT dispersion corrections,[40,41,92] are often computed using the classical scaling,

$$R_{\mathrm{vdW}}^{\mathrm{eff}} = \left(\alpha_1^{\mathrm{eff}}/\alpha_1^{\mathrm{free}}\right)^{1/3} R_{\mathrm{vdW}}^{\mathrm{free}}, \tag{34}$$

where $\alpha_1^{\mathrm{eff}}$ is derived from electron density[42] (see Sec. IV C). The QDO-based alternative in Eq. (30) avoids using empirical free-atom vdW radii[84] $R_{\mathrm{vdW}}^{\mathrm{free}}$,

$$R_{\mathrm{vdW}}^{\mathrm{eff}} = 2.54\ (\alpha_1^{\mathrm{eff}})^{1/7} \quad \text{a.u..} \tag{35}$$

This improves accuracy, as demonstrated for the TS method[16] and applied in the non-local MBD method (MBD-NL).[93] Moreover, Eq. (30) generalizes to heteronuclear vdW dimers

$$R_{\mathrm{e}}^{AB} = 2 \times 2.54 \left(\frac{\alpha_1^A + \alpha_1^B}{2}\right)^{1/7} \quad \text{a.u.,} \tag{36}$$

predicting their equilibrium distances within 2.5%.[16] These findings enable predictive, fully analytical interatomic vdW potentials, further discussed in Sec. VI D.

### B. Polarizability vs atomic volume

Understanding the relation between polarizability and atomic or molecular volume enables the development of simple approximations that circumvent costly *ab initio* response simulations. Classical electrostatics suggests a linear proportionality between dipole polarizability and volume, $\alpha_1 \propto V$, as discussed in Eq. (33). Depending on how the molecular volume is defined, this linear relation has been confirmed by several authors,[94,95] while others have suggested a superlinear dependence.[96] For free atoms, both linear and superlinear behaviors have been reported, depending on whether the electron shell is closed or open.[97] Notably, a linear relation between atomic polarizability and volume has been applied in the TS, XDM, and MBD methods.

In contrast, the QDO model provides a clear quantum-mechanical scaling law for polarizability as a function of volume.[15] For the QDO particle density $n_0(\mathbf{r}) = e^{-r^2/2\sigma^2}/(\sqrt{2\pi}\sigma)^3$, the effective volume is given by[15,17]

$$V = \int r^3\, n_0(\mathbf{r})\, d\mathbf{r} = \frac{16\sigma^3}{\sqrt{2\pi}}, \tag{37}$$

where $\sigma$ is the spread of the QDO from Eq. (5). Using this expression in combination with Eq. (6) yields scaling relations for the dipole, quadrupole, and octupole polarizabilities,

$$\begin{aligned} \alpha_1 &= \left(\frac{\pi}{16}\right)^{2/3} \frac{\mu q^2}{\hbar^2} V^{4/3}, \\ \alpha_2 &= \left(\frac{3\pi}{64}\right) \frac{\mu q^2}{\hbar^2} V^2, \\ \alpha_3 &= \frac{5}{4}\left(\frac{\pi}{16}\right)^{4/3} \frac{\mu q^2}{\hbar^2} V^{8/3}. \end{aligned} \tag{38}$$

These relations imply consistent scaling between different multipole polarizabilities

$$\frac{\mu q^2}{\hbar^2} \propto \frac{\alpha_1}{V^{4/3}} \propto \frac{\alpha_2}{V^2} \propto \frac{\alpha_3}{V^{8/3}}, \tag{39}$$

$$\alpha_1 \propto (\alpha_2)^{2/3} \propto (\alpha_3)^{1/2}, \tag{40}$$

consistent with the empirically observed relation in Eq. (32). These scaling laws are also in close agreement with the numerical observations $\alpha_1 \propto (\alpha_2)^{0.6} \propto (\alpha_3)^{0.44}$ reported in Ref. 98 for large molecular systems—fullerenes.

More recently, it has been shown that the fourth-power scaling between dipole polarizability and quantum-mechanical length scale holds universally across a wide range of one-electron model Hamiltonians.[81] This result is obtained by considering the Euclidean $\mathcal{L}^2$ norm $L_i$ of the position vector $(\mathbf{r} - \mathbf{R})$ as a natural length scale for quantum systems with ground-state wave function $\Psi_0(\mathbf{r})$,

$$L_i = \sqrt{\int (r_i - R_i)^2\, |\Psi_0(\mathbf{r})|^2\, d\mathbf{r}^N}, \tag{41}$$

where $\mathbf{R} = \langle \Psi_0 | \hat{\mathbf{r}} | \Psi_0 \rangle$ is the center of mass (coinciding with the nuclear position for atoms) and $N$ is the spatial dimensionality. In this framework, the polarizability along a principal axis is given by[81]

$$\alpha_1^{ii} = C_i (4\mu q^2/\hbar^2) L_i^4, \tag{42}$$

where $q$ and $\mu$ are the charge and mass of the quantum particle, and $C_i$ is the dimensionless constant. For the QDO model, $C_i = 1$ exactly, which serves as a reasonable approximation for other quantum systems as well.[81] The four-dimensional scaling law has also been further generalized[82] to many-electron atoms by accounting for electron correlation effects. This was accomplished by redefining atomic size using one- and two-electron reduced density matrices and introducing an effective number of valence electrons. For further details, we refer the reader to Ref. 82.

The universal relation (42) can be exploited in atom-in-molecule (AIM) schemes to refine predictions of molecular polarizabilities as follows:[81,82]

$$\alpha_1^{\mathrm{mol}} = \sum \alpha_1^{\mathrm{free}} \left(\frac{V_{\mathrm{eff}}}{V_{\mathrm{free}}}\right)^{4/3}, \tag{43}$$

where the summation is performed over atoms. The quantum-mechanical 4/3 scaling corrects the systematic overestimation of molecular polarizabilities found in classical linear scaling models.[81,82] It offers a straightforward refinement of existing vdW

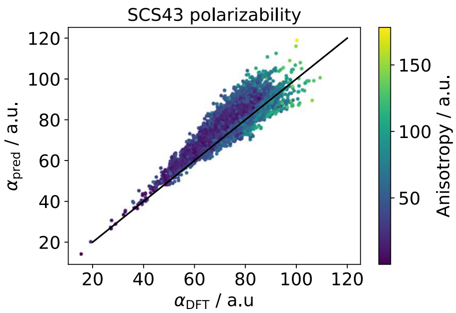


**FIG. 6.** Trace of dipole polarizability tensor predicted by the "SCS43" methods vs PBE0[99] reference values on the QM7–X dataset of small organic molecules.[100] The color of each point shows the anisotropy of the polarizability tensor. Reproduced with permission from S. Góger, M. R. Karimpour, and A. Tkatchenko, J. Chem. Theory Comput. **20**, 6621 (2024). Copyright 2024 American Chemical Society.

methods that use volume rescaling of $\alpha_1$, with minimal required modifications to electronic-structure codes.

Finally, short-range electrodynamic screening effects on local atomic polarizabilities (tensors) $\boldsymbol{\alpha}_j$ can be captured via a Dyson-like self-consistent screening (SCS) equation[7]

$$\boldsymbol{\alpha}_j^{\text{SCS}}(\omega) = \boldsymbol{\alpha}_j^0(\omega) - \boldsymbol{\alpha}_j^0(\omega)\sum_{k\neq j}\mathbf{T}\boldsymbol{\alpha}_k^{\text{SCS}}(\omega), \tag{44}$$

where $\mathbf{T}$ is the dipole–dipole interaction tensor and $j, k$ denote atomic indices. Combining the 4/3 scaling of AIM polarizabilities with the SCS approach yields the so-called SCS43 model[82] for molecular polarizabilities, relying only on the ground-state electron density. As illustrated in Fig. 6, this method reproduces isotropic molecular polarizabilities at the PBE0 level of theory.[99] Moreover, it eliminates the positive correlation between prediction errors and polarizability anisotropy found in local isotropic approaches such as Eq. (43). Potential improvements of the model include the incorporation of long-range charge transfer effects and refining the shape of the interaction potential.[82]

### C. QDO parameterization schemes

With only three parameters, the predictive power of the QDO model relies critically on how these parameters are chosen. A typical strategy involves using the closed-form expressions for the QDO response properties of Eqs. (6) and (9), which can be inverted to set the three parameters $\{q, \mu, \omega\}$ such that these properties are exactly reproduced. While six response properties are available in Eqs. (6) and (9), any three of them can be used to uniquely determine the QDO parameters.

For instance, Jones *et al.*[13] proposed using $\{\alpha_1, C_6, C_8\}$ to parameterize the QDO (JQDO scheme), yielding

$$q = \sqrt{\mu\omega^2\alpha_1}, \quad \omega = 4C_6/3\hbar\alpha_1^2, \quad \mu = 5\hbar C_6/\omega C_8. \tag{45}$$

Alternatively, prioritizing multipole polarizabilities, one can use $\alpha_1, \alpha_2, C_6$, leading to slightly different values for $\mu$, while $q$ and $\omega$ remain the same,[37]

$$q = \sqrt{\mu\omega^2\alpha_1}, \quad \omega = 4C_6/3\hbar\alpha_1^2, \quad \mu = 3\hbar\alpha_1/4\omega\alpha_2. \tag{46}$$

In contrast, the optimized QDO parameterization (OQDO scheme) uses only two properties, $\{\alpha_1, C_6\}$, as described in Ref. 18. A third constraint is introduced through a quantum-mechanical relation (31) between $\alpha_1$ and $R_{\text{vdW}}$, leading to a transcendental equation for $\mu\omega$, which is solved numerically for every species given only $\alpha_1$.[18] The remaining parameters are computed via $\omega = 4C_6/3\hbar\alpha_1^2$ and $q = \sqrt{\mu\omega^2\alpha_1}$. While OQDO and JQDO yield identical $\omega$, they differ in $\mu\omega$ and $q$. Tabulated OQDO parameters for the periodic table can be found in Ref. 18. The OQDO scheme works accurately for modeling spatial polarization distributions[18] and vdW-induced electron density shifts,[20] whereas JQDO provides expectedly more accurate higher-order dispersion coefficients.[18]

Another approach is the fixed-charge QDO (FQDO) scheme, which sets $q = 1$ a.u., assuming that drudons carry the same charge as electrons. The FQDO parameters are given by

$$q = 1, \quad \omega = 4C_6/3\hbar\alpha_1^2, \quad \mu = 1/\alpha_1\omega^2, \tag{47}$$

aligning with setting $q = 1$ a.u. in $\alpha_1 = q^2/\mu\omega^2$. This approach is implicitly used in the many-body dispersion (MBD) model.[7,19,23,101] While the MBD *energy* is invariant with respect to the choice of $q$ (given fixed $\{\alpha_1, C_6\}$), other properties of the MBD solution—such as charge polarization[19,20,101] or the dispersion–polarization energy[23] (see Sec. V E)—depend on all three QDO parameters.

For molecular simulations, where the QDO model is usually combined with a mean-field electronic structure method, effective parameters (polarizabilities and dispersion coefficients) for atom-based oscillators are often derived via atom-in-molecule approaches. These schemes typically exploit certain proportionality relations between the parameters and quantities available from atom-in-molecule partitioning methods.

As pointed out in Sec. IV B, the vdW(TS), XDM, and MBD frameworks rely on a linear relation between atomic volume and polarizability. Accordingly, the effective atomic polarizability $\alpha_1^{\text{eff}}$ is obtained by rescaling its *in vacuo* value by the ratio of the corresponding effective and free-atom volumes, obtained by Hirshfeld partitioning of the electron density,[42,102]

$$\alpha_1^{\text{eff}} = \alpha_1^{\text{free}}\left(\frac{V^{\text{eff}}}{V^{\text{free}}}\right). \tag{48}$$

In addition, with the evident inconsistency of the $\alpha_1 \propto V$ assumption with the QDO model (see Sec. IV B), this approach suffers from the known shortcomings of the standard Hirshfeld partitioning scheme. Chief among them, standard Hirshfeld analysis routinely underestimates the effect of charge transfer,[103] consequently overestimating polarizabilities for (fractionally) cationic species and underestimating them for (fractionally) anionic ones.

The improved, iterative Hirshfeld partitioning[104] allows us to alleviate many of these shortcomings. To the same end, Gould *et al.*[105] developed a fractionally ionic approach, interpolating between polarizabilities of two nearby ionic reference states based on the atom's effective charge. The MBD-NL method[93] further extends the MBD parameterization to accurately describe complex systems, including ionic as well as transition metal compounds, by employing a non-local VV10 polarizability functional.[106,107]

Effective atomic polarizabilities can also be obtained based on the density matrix in an atom-centered basis set. Here, the linear correlation between the (effective) polarizability of an atom and the atom-projected trace of the density matrix is used in a rescaling scheme similar to Eq. (48). This alternative enables the parameterization of the QDO model to density-free approaches, including semi-empirical methods.[108,109] In the MBD context, Bereau and von Lilienfeld[110] proposed a simple approach using Voronoi tessellation of the molecular geometry, neglecting electronic effects altogether but providing an effective parameterization for molecular mechanics.

## D. Polarization potential

As discussed in Sec. III H, the QDO model provides a robust framework for describing the polarization response of atoms and molecules in the presence of external electric fields. Building on this foundation, Góger *et al.*[18] recently demonstrated that the QDO model can also accurately reproduce atomic polarization potentials, a key quantity characterizing field-induced electron density distortions.

In this context, the polarization potential $V_{\rm pol}(\mathbf{r})$ is defined[18] as the change in the electrostatic potential of a system resulting from the polarization of its charge density by an external electric field $\boldsymbol{\mathcal{E}}$,

$$V_{\rm pol}(\mathbf{r}) = -\frac{1}{4\pi\varepsilon_0}\int \frac{\rho_{\mathcal{E}}(\mathbf{r}') - \rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\, \mathrm{d}^3\mathbf{r}', \tag{49}$$

where $\rho(\mathbf{r})$ denotes the unperturbed ground-state electron density and $\rho_{\mathcal{E}}(\mathbf{r})$ is the electron density polarized by the external field. This definition of the polarization potential can be conceptually linked to the interaction energy between two subsystems when the action of one is approximated by a static external field, providing a physically grounded interpretation of polarization potentials.[18]

Within the QDO model framework, the polarization potential of Eq. (49) can be treated analytically, yielding

$$V_{\rm pol}^{\rm QDO}(\mathbf{r}) = \frac{-q}{4\pi\varepsilon_0}\left(\frac{\mathrm{erf}(\tilde{r}/\sigma\sqrt{2})}{\tilde{r}} - \frac{\mathrm{erf}(r/\sigma\sqrt{2})}{r}\right), \tag{50}$$

where $\tilde{\mathbf{r}} = \mathbf{r} - \alpha_1\boldsymbol{\mathcal{E}}/q$ is the field-shifted oscillator coordinate, and $\sigma$ is the natural QDO length of Eq. (5). The closed-form expression of Eq. (50) provides valuable insight into the spatial variation of polarization potentials and allows for efficient evaluation across different systems.

Figure 7 compares the polarization potentials $V_{\rm pol}$ computed from DFT-PBE0 for H and Ar atoms with those obtained analytically from Eq. (50) using the three QDO parameterization schemes discussed in Sec. IV C: the fixed-charge QDO (FQDO), the Jones QDO (JQDO), and the optimized QDO (OQDO). All models were parameterized using high-level reference data for $\alpha_1$, $C_6$, and, when available, dipole–quadrupole dispersion coefficients $C_8$.[15,87,88,111–113]

Among these three parameterization schemes, the OQDO model shows the best overall agreement with the DFT-PBE0 reference, achieving a normalized root-mean-square error (RMSE) of 8.9% across a dataset of 21 atoms. In comparison, the JQDO and FQDO models yield higher RMSEs of 13.2% and 15.4%, respectively.[18] Notably, the OQDO formulation performs well not only for light atoms but also for heavier, many-electron systems, including noble gases and alkali metals, demonstrating its broad applicability and high accuracy.

The ability of the OQDO model to reproduce non-linear $V_{\rm pol}(\mathbf{r})$ profiles from DFT calculations without additional adjustments emphasizes the critical importance of optimal parameter mapping in QDO-based models. This result reinforces the value of the QDO approach as a predictive and physically grounded method for modeling polarization effects across the periodic table.

## E. QDO model and the exchange-hole dipole method

An alternative perspective on vdW dispersion, distinct from conventional perturbation theory,[4] was proposed in the exchange-hole dipole moment (XDM) model by Becke and Johnson.[114,115] In what follows, we will demonstrate the connection between the XDM and QDO models for the vdW dispersion energy.

Within the XDM model, dispersion interactions arise from the coupling between electronic multipoles associated with a moving electron and its accompanying exchange-correlation hole, which is approximated by the exchange hole. The exchange hole describes the instantaneous depletion of probability density for finding a same-spin electron near a reference electron due to Pauli exclusion.[114,115] While the electron–exchange-hole pair is neutral overall, it possesses a non-zero dipole moment, $\mathbf{d}_X(\mathbf{r})$, due to the aspherical shape of the



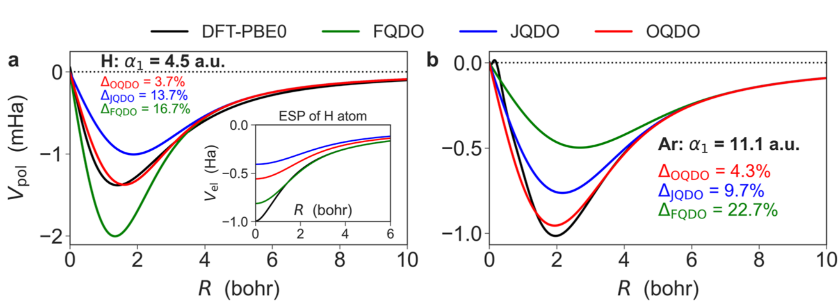


**FIG. 7.** Polarization potential $V_{\rm pol}(\mathbf{r})$ calculated with DFT-PBE0 and different QDO parameterizations for (a) hydrogen and (b) argon atoms. In all cases, the direction along the applied field was chosen for the plots. The numerical values of the normalized root-mean-squared error ($\Delta$) are displayed for the three QDO flavors. For the hydrogen atom, the unperturbed electrostatic potential (ESP) $V_{\rm el}(\mathbf{r})$ is shown as an inset, indicating that the QDO model captures the response of atomic electron density but not the ESP itself. Adapted from Góger *et al.*, J. Phys. Chem. Lett. **14**, 6217 (2023). Copyright 2023 American Chemical Society.

exchange hole. The resulting XDM dipole–dipole dispersion energy is given by[114]

$$E_{\text{dip-dip}} = -\frac{C_6}{R^6} = -\frac{1}{2}\langle d_X^2\rangle\frac{\alpha_1}{R^6}, \tag{51}$$

where

$$\langle d_X^2\rangle = \int n(\mathbf{r})d_X^2(\mathbf{r})d\mathbf{r}, \tag{52}$$

is the expectation value of the squared exchange-hole dipole moment magnitude.

Since its introduction, the XDM model has undergone important theoretical refinements. Contributions from Ángyán,[116,117] Ayers,[118] and Hesselmann[119] have provided deeper insights into the model's foundation. These studies revealed that the expectation value $\langle \mathbf{r}\cdot\mathbf{d}_X\rangle$ holds a more fundamental role, entering the expressions for multipole moments, than $\langle d_X^2\rangle$, which XDM traditionally uses. Although XDM does not emerge from a strictly rigorous derivation, it offers a physically sound and computationally efficient framework for modeling dispersion interactions.[92] Remarkably, the XDM approach becomes exact in the case of QDOs,[120] as demonstrated below.

For spherically symmetric electron densities, Becke and Johnson showed that[114]

$$\langle d_X^2\rangle = \langle p^2\rangle = q^2\int r^2 n(\mathbf{r})d\mathbf{r}, \tag{53}$$

where $\langle p^2\rangle$ is the expectation value of the squared dipole moment operator $\mathbf{p} = \sum q_i\mathbf{r}_i$. In the QDO model, this becomes

$$\langle d_X^2\rangle = \langle p^2\rangle = 3q^2\sigma^2 = \frac{3}{2}\frac{q^2\hbar}{\mu\omega} = \frac{3}{2}\alpha_1\hbar\omega, \tag{54}$$

where we used $\sigma$ of Eq. (5). Substituting this into Eq. (51) yields

$$E_{\text{dip-dip}} = -\frac{3}{4}\frac{\alpha_1^2\hbar\omega}{R^6}, \tag{55}$$

which matches the QDO result from perturbation theory, see Eq. (9).

This result can also be derived from the theory of insulating states (TIS) of Resta,[121] starting from the general sum rule,

$$\frac{1}{N_e}\int_0^\infty du\ \operatorname{Im}\alpha_{\beta\gamma}(u) = \frac{\pi q^2}{\hbar}\langle r_\beta r_\gamma\rangle_c\ . \tag{56}$$

Here, $N_e$ is the number of electrons, and $\alpha_{\beta\gamma}(u)$ is the frequency-dependent polarizability tensor. The localization tensor in TIS is given by[121]

$$\langle r_\beta r_\gamma\rangle_c = \frac{\langle\Psi|\hat{R}_\beta\hat{R}_\gamma|\Psi\rangle - \langle\Psi|\hat{R}_\beta|\Psi\rangle\langle\Psi|\hat{R}_\gamma|\Psi\rangle}{N_e}, \tag{57}$$

where $\hat{\mathbf{R}} = \sum_{i=1}^{N_e}\mathbf{r}_i$ and $\Psi$ is the many-electron ground-state wave function. Assuming spherical symmetry, we consider the diagonal elements $\alpha_{\beta\beta}(u) = \alpha_1(u)$ and $q^2\langle r_\beta r_\beta\rangle_c = \langle p_{\beta\beta}^2\rangle/N_e = \frac{1}{3}\langle p^2\rangle/N_e$ in Eq. (56). Noting that the QDO model satisfies[6]

$$\int_0^\infty du\ \operatorname{Im}\alpha_1(u) = \frac{\pi}{2}\frac{q^2}{\mu\omega}, \tag{58}$$

we recover $\langle d_X^2\rangle = \langle p^2\rangle = \frac{3}{2}\frac{q^2\hbar}{\mu\omega} = \frac{3}{2}\alpha_1\hbar\omega$, consistent with Eq. (54).

Equation (56) highlights the fluctuation–dissipation connection: its left-hand side represents dissipation (via polarizability), while the right-hand side reflects fluctuations (via the exchange-hole dipole). As shown in Refs. 116–118, the QDO model elegantly bridges perturbation theory and XDM within a unified formalism through this fluctuation–dissipation link.

## V. BEYOND PERTURBATIVE TREATMENT OF THE COUPLED QDO PROBLEM

In this section, we review the existing theoretical frameworks that enable a (fully) non-perturbative treatment of the coupled QDO Hamiltonian, including both dipolar and full Coulomb interactions. While perturbation theory provides valuable physical insight and computational efficiency for weakly interacting systems, it becomes inadequate in regimes involving strong coupling, significant overlap of electronic clouds, or collective many-body effects. To address these challenges, a variety of non-perturbative approaches have been developed, ranging from quantum Monte Carlo techniques and (variational) quantum chemistry methods to the many-body dispersion formalism and path integral molecular dynamics. These methods allow for a more accurate and often more physically transparent description of dispersion and polarization phenomena in systems where perturbative methods fall short.

### A. Quantum Monte-Carlo framework for QDOs

Solving the full Coulomb-coupled QDO Hamiltonian (2) requires numerical techniques, as no analytical solution exists for this many-body quantum problem. Among the most effective approaches for this task are Quantum Monte Carlo (QMC) methods, which provide a scalable and statistically robust tool to simulate correlated quantum systems.

A dedicated QMC framework tailored to QDO systems was recently developed by Ditte *et al.*[28,122,123] and is available on request within the QMeCha package.[124] This software offers versatile tools for performing QMC simulations using a range of wave function ansätze. It supports not only model systems of Coulomb-coupled QDOs but also hybrid scenarios where electronic systems (atoms and molecules) are embedded in a polarizable QDO environment.[122,123] This enables the explicit treatment of many-body electronic-polarizable interactions, including all correlation effects.[122,123]

The QMeCha framework thereby opens a wide range of applications. It allows fully quantum mechanical embedding of solutes in polarizable solvents, providing a powerful alternative to continuum or classical polarizable models.[122,123] Moreover, it facilitates the study of strongly Coulomb-bound QDO dimers—an idealized setting that can serve as a minimal quantum model for covalent bond formation and dissociation.[28]

### B. Quantum chemistry with QDO model

Quantum-chemical methods provide an alternative route for solving the Coulomb-coupled QDO problem, offering valuable insights into the nature of quantum fluctuations in neutral atomic systems. Early studies in this direction explored both one-dimensional[125] and three-dimensional[14] QDO dimers as model systems to simulate vdW dispersion interactions arising from quantum charge-density fluctuations.

These investigations assessed the performance of various quantum-chemical approaches—such as the random phase approximation (RPA) and Møller–Plesset perturbation theory (MPPT)—against highly accurate full configuration interaction (FCI) benchmarks. It was revealed that, both within dipole approximation[125] and under the full Coulomb interaction,[14] RPA tends to significantly underestimate mid-range interactions. This shortcoming arises from the lack of relaxation contributions that would typically be captured by single excitations, missing in the standard RPA framework. In contrast, higher-order MPPT with full Coulomb interaction avoids this limitation by capturing the missing contributions more effectively.

However, in these studies, the atomic polarizabilities were considered immutable,[13] neglecting their potential variations due to the interactions between atomic fragments. Incorporating self-consistent polarizability parameterization, as performed in the exact dipole-coupled MBD method (which is essentially an "RPA in disguise"[126]), leads to significantly improved accuracy (see Sec. V D).

The aforementioned quantum-chemical explorations not only provide benchmarks for theoretical modeling but also suggest future directions. For example, insights from these studies can be used to guide the development of improved damping functions for many-body dispersion corrections. Such refinements could further bridge the gap between first-principles methods and physically grounded analytical models.

### C. Variational treatment of coupled QDOs

An alternative way of numerical solution for energies of coupled QDOs is to use the variational approach.[127,128] In this framework, the wave function of the system is constructed from a basis of non-interacting QDO eigenfunctions,

$$\psi_{k\ell m}(r,\theta,\phi) = N_{k\ell}\, r^{\ell} e^{-\nu r^2} L_k^{(\ell+\frac{1}{2})}(2\nu r^2)\, Y_\ell^m(\theta,\phi), \tag{59}$$

where $N_{k\ell}$ is the normalization constant, $L_k^{(\ell+\frac{1}{2})}(x)$ are the generalized Laguerre polynomials, $Y_\ell^m(\theta,\phi)$ are the spherical harmonic functions, $\nu = \mu\omega/2\hbar$, and $k, \ell, m$ are the quantum numbers. In Ref. 127, which considered systems comprising 2–4 interacting QDOs, the wave function of each oscillator $i$ is expanded as a linear combination of these basis functions,

$$\Psi_i = \sum_n c_{i,n}\psi_{i,n}, \tag{60}$$

where $c_{i,n}$ are the variational coefficients to be determined. The above-mentioned expansion allows a flexible yet systematically improvable description of each oscillator's behavior under coupling.

The Hamiltonian of two interacting QDOs is given by Eq. (2). For a general multi-oscillator system, the total Hamiltonian becomes

$$\hat{H} = \sum_i \hat{H}_{0,i} + \sum_{i\neq j}\left(\hat{U}_{ij}^{\rm dn} + \frac{1}{2}\hat{U}_{ij}^{\rm dd} + \frac{1}{2}\hat{U}_{ij}^{\rm nn}\right), \tag{61}$$

where $\hat{H}_{0,i}$ is the single-oscillator Hamiltonian of Eq. (1), and the three potential terms in parentheses describe the Coulomb interactions between a drudon and neighboring pseudo-nucleus, between two drudons, and between two pseudo-nuclei, respectively.

The total energy of the system can be expressed as a sum of contributions from single-drudon, two-drudon, and pseudo-nuclei interaction terms,

$$E = \sum_i \left\langle \Psi_i \middle| \hat{H}_{0,i} + \sum_{i\neq j}\hat{U}_{ij}^{\rm dn} \middle| \Psi_i \right\rangle + \sum_{i\neq j}\left\langle \Psi_i\Psi_j \middle| \frac{1}{2}\hat{U}_{ij}^{\rm dd} \middle| \Psi_i\Psi_j \right\rangle + \sum_{i\neq j}\frac{1}{2}\hat{U}_{ij}^{\rm nn}. \tag{62}$$

The energy can then be minimized with respect to the basis-set expansion coefficients, $c_{i,n}$, using the method of Lagrange multipliers.[127]

Once the optimized wave functions of Eq. (60) are obtained, the corresponding electron densities can be constructed, which enables direct coupling to density-based dispersion correction methods, such as the TS, XDM, and MBD approaches. As such, this variational framework allows for detailed investigations of how the $C_6$ dispersion coefficients and interaction energies obtained with the listed methods vary when applied to interacting model systems of 2–4 QDOs.[127]

Moreover, the variational treatment of coupled QDOs provides a rigorous way to probe the role of many-body effects beyond pairwise interactions. It allows for the assessment of the relative importance of three-body and higher-order contributions to the total dispersion energy. In addition, it offers insight into the sensitivity of the SCS procedures employed in methods like MBD, particularly with respect to variations in the input static polarizabilities.[127] Overall, the considered approach offers a powerful platform for systematically testing and improving dispersion models within a fully quantum mechanical framework.

### D. Many-body dispersion method

Within the dipole approximation, the Hamiltonian of $N$ coupled QDOs can be written as[7]

$$\hat{H}_{\rm MBD} = \sum_A\left[-\frac{\hbar^2}{2}\nabla_{\xi_A}^2 + \frac{1}{2}\omega_A^2\xi_A^2\right] + \frac{1}{2}\sum_{A\neq B}\omega_A\omega_B\sqrt{\alpha_{1,A}\alpha_{1,B}}\,\boldsymbol{\xi}_A\mathbf{T}_{AB}\boldsymbol{\xi}_B, \tag{63}$$

where $\boldsymbol{\xi}_A = \sqrt{m_A}(\mathbf{r}_A - \mathbf{R}_A)$ are mass-weighted charge displacements, and $\mathbf{T}_{AB}$ is the dipole interaction tensor. This bilinear Hamiltonian can be solved numerically exactly via eigenmode transformation, yielding $3N$ uncoupled 1D quantum harmonic oscillators in the basis of collective oscillation modes with frequencies $\tilde{\omega}_k$, which underpin the MBD method.[7] The MBD energy, representing the zero-point energy change due to dipole interactions, is

$$E_{\rm MBD} = \sum_{k=1}^{3N}\frac{\hbar\tilde{\omega}_k}{2} - \sum_{A=1}^{N}\frac{3\hbar\omega_A}{2}. \tag{64}$$

This formula is equivalent to the ACFDT-RPA (adiabatic connection fluctuation–dissipation theorem) correlation energy in the dipole approximation,[129] capturing effects up to infinite order in perturbation theory and up to order $N$ in many-body expansion.[126] The equivalence with RPA is easily understood by considering the self-consistent electrodynamic coupling between point-like atomic dipoles at real frequency $\omega$,

$$\boldsymbol{\xi}_A = -\frac{\alpha_A\omega_A^2}{\omega_A^2 - \omega^2}\sum_{B\neq A}\boldsymbol{T}_{AB}\boldsymbol{\xi}_B. \tag{65}$$

where we used the QDO dynamical polarizability $\alpha_A(\omega) = \alpha_A\omega_A^2/(\omega_A^2 - \omega^2)$.

Recasting this equation into an eigenvalue problem

$$\omega^2\boldsymbol{\xi}_A = \sum_B \left[(1-\delta_{AB})\alpha_A\omega_A^2\boldsymbol{T}_{AB} + \delta_{AB}\omega_B^2\right]\boldsymbol{\xi}_B, \tag{66}$$

one finds that the self-sustained collective dipolar modes correspond to vanishing determinant solutions of the above-mentioned equation. This operation determines the frequency $\omega$ of the collective RPA modes, which are mathematically equivalent to MBD eigenmodes.[126,129]

The MBD method developments are implemented in the LIBMBD library by Hermann *et al.*,[101] which features a PYTHON API and interfaces with major electronic-structure codes. Alternative MBD formulations, such as fractionally ionic MBD[105] and unified MBD,[130] are available outside LIBMBD. Recent advancements include neural network models for predicting Hirshfeld volume ratios[131] and stochastic estimators for the MBD Hamiltonian trace,[132] enabling linear-scaling, density-free methods. In addition, MBD generalizations with quadrupole coupling have been explored.[30–32] Furthermore, MBD has also been integrated within the DFT-D4 method for dispersion corrections to account for many-body effects.[133] Finally, the combination of SAPT and MBD within the extended SAPT+MBD (XSAPT+MBD) methodology[134,135] generalizes traditional SAPT beyond dimers to extended many-body systems—such as protein–ligand complexes—offering an accurate yet computationally efficient description of noncovalent interactions.

One of the main advantages of MBD compared to other post-DFT vdW methods is that beyond providing a numerical estimate of many-body dispersion energy, the MBD method offers a physically grounded description of quantum electronic density fluctuations relative to the DFT reference. The density of MBD exciton states exhibits qualitatively similar features with the Bethe–Salpeter-derived electronic excitons in small molecules,[136] motivating new conceptual frameworks. A recent second-quantized MBD formalism (SQ-MBD)[137] makes it possible to describe MBD excitons directly in terms of local QDO excited states, bridging local and collective quantum fluctuations. SQ-MBD naturally enables projections of observables obtained from the MBD model—such as interaction energies, transition multipoles, or polarizabilities—onto coarse-grained representations based on atoms or large structural motifs. Moreover, the SQ-MBD framework opens the door to applying quantum-information conceptual tools for analyzing correlations and (non)separability of quantum electronic density fluctuations among fragments in a given molecular complex. These insights pave the way toward extending MBD to ever-larger and intricate systems while capturing complex dispersion and polarization phenomena with high accuracy.

### E. Beyond-RPA many-body dispersion framework

In this section, we present an efficient method for incorporating beyond-RPA effects into the MBD framework by evaluating the first-order correction due to full Coulomb coupling in a system of dipole-coupled QDOs. Analogous to terminology in quantum chemistry, we refer to this correction as *Dipole-coupled Coulomb Singles* (DCS). The DCS energy is defined as

$$E_{\text{DCS}} = E_{\text{C}}[\Psi_{\text{DC}}] - E_{\text{C}}[\Psi_0] - E_{\text{dip}}[\Psi_{\text{DC}}], \tag{67}$$

with

$$E_{\text{X}}[\Psi] = \langle\Psi|\sum_{A<B}\hat{V}_{AB}^{(\text{X})}|\Psi\rangle, \tag{68}$$

where X denotes full Coulomb (C) or dipolar (dip) interactions between oscillators. $\Psi_{\text{DC}}$ is the wave function of the dipole-coupled state, and $\Psi_0$ is the wave function of the reference system of non-interacting QDOs. Note that $E_{\text{dip}}[\Psi_0] = 0$.

Separating the classical, $J[\rho]$, and correlation, $E_{\text{corr}}[\Psi]$, components of the total energy, the DCS contribution can be written as

$$E_{\text{DCS}} = J[\rho_{\text{DC}}] - J[\rho_0] + E_{\text{corr}}[\Psi_{\text{DC}}] - E_{\text{corr}}^{(\text{dip})}[\Psi_{\text{DC}}], \tag{69}$$

where we introduced the charge density $\rho$ corresponding to the wave function $\Psi$. The first two terms represent the difference in the electrostatic energy between the dipolar-correlated and uncorrelated states, illustrating the connection between DCS and dispersion–polarization.[23] The total DCS (interaction) energy also contains a correction for the beyond-dipolar correlation energy (second pair of terms); however, dispersion–polarization is the dominating effect. Overall, DCS acts as a first-order perturbative correction to the non-perturbative MBD solution, thus blending both methodologies.

The key to obtaining the Coulomb energy of the dipole-coupled (=MBD) state is to evaluate the expectation value of the individual two-particle Coulomb operator, which we can write as

$$\hat{V}_{AB}^{(\text{C})} = \frac{q_A\,q_B}{\|\mathbf{r}_A - \mathbf{r}_B\|} = \frac{2\,q_A\,q_B}{\sqrt{\pi}}\int_0^\infty e^{-\mathbf{r}_{AB}^{\text{T}}\mathbf{U}_2\,\mathbf{r}_{AB}}\,du, \tag{70}$$

with

$$\mathbf{r}_{AB} = \begin{pmatrix}\mathbf{r}_A\\ \mathbf{r}_B\end{pmatrix} \quad\text{and}\quad \mathbf{U}_2 = u^2\begin{pmatrix}\mathbb{1}_3 & -\mathbb{1}_3\\ -\mathbb{1}_3 & \mathbb{1}_3\end{pmatrix}. \tag{71}$$

In the Appendix, we present a general approach for calculating expectation values over the MBD wave function. Following Eq. (A8) with $|\mathcal{X}| = |\{A,B\}| = 2$ and $\mathcal{X} = \{A,B\}$, denoted as $AB$ (a set of two QDOs), we get for the expectation value of the Coulomb operator

$$V_{AB}^{(\text{C})} = \frac{\sqrt{\det\{\boldsymbol{\Gamma}_{AB}\}}}{\pi^3}\int\int_0^\infty \exp\big(-\mathbf{r}_{AB}^{\text{T}}\mathbf{U}_2\,\mathbf{r}_{AB} - \boldsymbol{\xi}_{AB}^{\text{T}}\boldsymbol{\Gamma}_{AB}\boldsymbol{\xi}_{AB}\big)\,du\,d\boldsymbol{\xi}_{AB}. \tag{72}$$

After inserting the mass-weighted charge displacements, $\boldsymbol{\xi}_{AB} = \sqrt{m_A}(\mathbf{r}_A - \mathbf{R}_A) \oplus \sqrt{m_B}(\mathbf{r}_B - \mathbf{R}_B)$, and further simplifications, we arrive at

$$V_{AB}^{(\text{C})} = \frac{2\,q_A\,q_B}{\sqrt{\pi}}\int_0^\infty \sqrt{\frac{\det\{\boldsymbol{\Gamma}_{AB}\}}{\det\{\boldsymbol{\Gamma}_{AB}+\mathbf{U}_2\}}} \times e^{-\mathbf{R}_{AB}^{\text{T}}(\boldsymbol{\Gamma}_{AB}-\boldsymbol{\Gamma}_{AB}^{\text{T}}(\boldsymbol{\Gamma}_{AB}+\mathbf{U}_2)^{-1}\boldsymbol{\Gamma}_{AB})\mathbf{R}_{AB}}\,du, \tag{73}$$

where $\mathbf{R}_{AB} = \mathbf{R}_A \oplus \mathbf{R}_B$ is defined analogously to $\mathbf{r}_{AB}$ in Eq. (71) but for the positions of the centers of the QDOs instead of the corresponding Drude particles. $\boldsymbol{\Gamma}_{AB} = \boldsymbol{\Omega}_{\mathcal{X}\mathcal{X}} - \boldsymbol{\Omega}_{\mathcal{X}\bar{\mathcal{X}}}\,\boldsymbol{\Omega}_{\bar{\mathcal{X}}\bar{\mathcal{X}}}^{-1}\,\boldsymbol{\Omega}_{\bar{\mathcal{X}}\mathcal{X}}$ and $\boldsymbol{\Omega}$ is given by $\mathbf{C}\,\text{diag}\{\tilde{\omega}_k\}\,\mathbf{C}^{\text{T}}$, with $\mathbf{C}$ as the eigenmodes of the MBD

Hamiltonian (see Sec. V D). The submatrices of $\mathbf{\Omega}$ entering $\mathbf{\Gamma}_{AB}$, finally, follow the definitions in the Appendix.

So, overall, the pairwise Coulomb interaction for the dipole-coupled state is fully defined by the Drude charges $q_A$ and the normal mode transformation matrix, $\mathbf{C}$, and frequencies, $\tilde{\omega}$, obtained from solving the MBD model. The total Coulomb energy, $E_C[\Psi_C]$, is then found by simply adding all pairwise interactions. For the non-interacting state, we have $\mathbf{C} = \mathbb{1}$ and, consequently, $\mathbf{\Gamma}_{AB}$ is equal to a diagonal matrix of the non-interacting QDO frequencies, $\omega_k$. The remaining term from Eq. (67), $E_{\text{dip}}[\Psi_{\text{DC}}]$, is obtained as a scaled trace over the dipole–dipole interaction tensor represented in the basis of the dipole-coupled state,

$$E_{\text{dip}}[\Psi_{\text{DC}}] = \frac{1}{4}\,\text{Tr}\{\text{diag}\{\tilde{\omega}_k^{-1}\}\,\mathbf{W}^{\text{T}}\mathbf{T}\,\mathbf{W}\}, \tag{74}$$

where

$$\mathbf{W} = \text{diag}\{\omega_k\sqrt{\alpha_{1,k}}\}\,\mathbf{C}.$$

In a 3D-isotropic vacuum, it can be shown that the total DCS contribution approaches zero due to symmetry. However, with the presence of a polarizable environment breaking 3D-symmetry of vacuum (e.g., by introducing a third oscillator), the first-order Coulomb interaction between dipolar quantum fluctuations increases and shows a leading-order repulsive term. The asymptotic behavior and trends in the DCS energy are best exemplified by considering a model system of two QDOs in a restricted, lower-dimensional space, which mimics the presence of a polarizable environment (perfectly) screening quantum fluctuations along a given dimension. For a pair of QDOs in reduced dimensions, one generally finds the leading-order term to follow $c\alpha_2\hbar\omega\,R^{-5}$, where $c$ is 2, 3/2, and 3 for QDOs in 1D, 2D, and quasi-3D, respectively.[22]

This result, together with the analysis in Eq. (69), allows us to associate the leading-order repulsive term with the interaction of dispersion-induced static quadrupole moments and the field of dipolar quantum fluctuations.[22,23] The results for the model system further illustrate how the DCS contribution grows with increasing environment complexity and quadrupolar polarizability, $\alpha_2$, of the system. Practical application to realistic molecular systems confirms this behavior, showing negligible DCS contributions in the case of small molecular dimers and an increasing significance in large supramolecular complexes or non-covalent interactions within metallic carbon nanotubes.[23] The resulting long-range repulsive nature of the DCS interaction energy considerably reduces attraction in realistic systems, which might hold the key to understanding some of the puzzling observations for non-covalent interactions under confinement.

Furthermore, rigorous quantitative investigation of dispersion–polarization within the QDO model and the DCS contribution in practically relevant systems is subject to ongoing investigations. Current research in this regard addresses a generally applicable full parameterization for QDOs representing realistic molecules and materials (cf. Sec. IV C), as well as their coupling to suitable electronic structure methods like DFT.

### F. Path-integral molecular dynamics for QDOs

In this section, we introduce the path-integral formulation for the statistical mechanics of quantum systems[2,138] as a versatile, non-perturbative approach to solving the coarse-grained electronic structure of QDOs.[139] Central to this approach is the connection between the quantum time-evolution operator (propagator) and the statistical-mechanical partition function.

The propagator $K(x,t;\ x_0,t_0)$ describes the probability amplitude for a quantum particle to evolve from position $x_0$ at time $t_0$ to $x$ at time $t$. It is expressed as

$$K(x,t;x_0,t_0) = \langle x|\exp\left(-\frac{i}{\hbar}\hat{H}(t-t_0)\right)|x_0\rangle, \tag{75}$$

where $\hat{H}$ is the Hamiltonian of interacting QDOs. The propagator can be represented as a path integral through time slicing, wherein the evolution is divided into small intervals. Each path in a resulting sum is weighted by $e^{iS/\hbar}$, where $S$ is the action.[140] In contrast, the canonical partition function, $Z$, characterizing a system in thermal equilibrium at a temperature $T$, is given as the trace of the density matrix by $Z = \text{Tr}\left(e^{-\beta\hat{H}}\right)$, with $\beta = 1/k_BT$.[37]

The connection between the propagator and the partition function emerges under rotation of the time axis in the complex plane $t \to -i\hbar\beta = -i\tau$, yielding an imaginary time proportional to inverse temperature. This transformation, therefore, maps the conventional time propagator into a thermal counterpart according to

$$e^{-\frac{i}{\hbar}\hat{H}t} \to e^{-\beta\hat{H}}, \tag{76}$$

describing a thermal evolution of the system over an imaginary time interval $\hbar\beta$.[2]

Exploiting the analogy with real-time evolution further allows the partition function to be recast as a path integral over imaginary time, where each path is now weighted by $e^{-S_E/\hbar}$, with $S_E$ being the so-called Euclidean action.[140] This includes all possible configurations, not just small perturbations around a reference solution, enabling a fully non-perturbative treatment.

To compute observables, the imaginary time $\tau$ is discretized into $M$ intervals, approximating a continuous path $x(\tau)$ by $M$ discrete points.[141] Applied to QDO, this transforms the partition function into a polymer chain of $M$ beads connected by harmonic springs.[142] Each bead corresponds to an imaginary time slice and represents a configuration of the quantum particle. This mapping establishes an approximate isomorphism between the quantum partition function of a QDO and that of the classical chain,[143] which becomes exact in the large $M$ limit. Finally, the trace operation in the quantum partition function,

$$Z = \int dx\,\langle x|e^{-\beta\hat{H}}|x\rangle, \tag{77}$$

imposes periodic boundary conditions $x(0) = x(\beta\hbar)$, requiring that the chain close.

Molecular dynamics (MD) can then be used to sample the QDO partition function for this classically mapped chain system.[144] For a system of $N$ QDOs, this approach scales as $M \times N\log N$.[145] However, standard recipes for computing ground state observables like energy and pressure from the partition function typically exhibit large variances and slow convergence. Hence, the QDO-specialized low-variance estimators for these quantities have been developed.[29,146]

It is important to stress that MD here serves solely as a statistical sampling tool for the QDO partition function and does not represent real-time quantum dynamics. Nevertheless, the method can be incorporated into physical dynamical simulations where nuclear motion evolves adiabatically on the ground-state surface of the Drude particles.[12,147]

This is achieved via two-temperature molecular dynamics, wherein the "hot and fast" QDO degrees of freedom are maintained at artificially elevated temperatures, while "cold and slow" nuclear motions evolve at physically realistic temperatures. This adiabatic separation enables the path integral sampling to converge sufficiently well so as to generate an accurate average force on the nuclei, provided careful thermostatting[39] and staging transformations[148] are implemented.

Together, these approaches open a direct route for QDO methods in practical materials simulation, where dynamical trajectories are driven by forces obtained on-the-fly from the coarse-grained QDO electronic structure without invoking any parameterized potentials.

## VI. APPLICATIONS OF THE QDO MODEL IN ATOMISTIC SIMULATIONS

Due to their ability to capture a broad range of interactions with computational efficiency, the QDO-based models have found widespread use across diverse systems and length scales. In this section, we highlight selected applications of QDO-based models in atomistic simulations to showcase their versatility. Rather than providing an exhaustive account, we focus on several illustrative examples, including the modeling of excess electrons in condensed matter, coarse-grained descriptions of water, practical QDO-based methods for van der Waals dispersion, and the construction of universal interatomic vdW potentials. These examples demonstrate how the QDO framework offers both conceptual clarity and practical utility in atomistic modeling. A more comprehensive overview of applications can be found in the references cited throughout this review and the studies therein.

### A. Interactions between excess electrons and molecules

Long-range interactions between an excess electron and a molecule or a cluster can lead to the formation of bound non-valence anions. In such anions, the electron is bound primarily through long-range electrostatic, dispersion, and polarization interactions. Such systems are challenging to treat using *ab initio* electronic structure methods due to the crucial role of electron correlation effects, requiring large basis sets with multiple diffuse functions to achieve energy convergence.

Due to the highly diffuse nature of the excess electron in non-valence anions, they can be well described using one-electron model Hamiltonians. In 2001, Wang and Jordan[10] introduced a model based on QDOs for describing the polarization and dispersion interactions between the excess electron and the valence electrons of the system. This model incorporated the electron–QDO interactions (in the form of charge–dipole coupling) as well as the dipole–dipole coupling between the QDOs to account for dispersion interactions. The electron binding energies were obtained from configuration interaction (CI) calculations employing the basis set of Gaussian-type orbitals (GTOs) for the excess electron as well as the ground and the first excited state eigenfunctions of the QDOs.[10]

Initial studies focused on electron binding to HCN and HNC molecules,[10] and the model was shortly afterward extended to treat excess electron binding to water clusters.[149–152] It was demonstrated that the one-electron QDO-based model Hamiltonian yields electron binding energies in excellent agreement with results from coupled-cluster calculations using large diffuse GTO basis sets, but only at a fraction of the computational cost. Later, Sommerfeld *et al.*[153] demonstrated that, by making an adiabatic approximation to the interaction between the excess electron and the QDOs, the molecular response to the electron can be described through polarization potentials.

### B. Coarse-grained modeling of water

A fundamental challenge in developing advanced empirical potentials for materials simulation is ensuring that they are transferable. That is, they perform well across a wide range of thermodynamic environments (temperature, pressure, and density) and phases (liquid, vapor, and solid), including those far from the parameterization conditions. Water is widely acknowledged as an exceptionally demanding test case in this regard. Despite its structural simplicity, water exhibits diverse polymorphism[154] and anomalous behavior,[155] which have consistently defied proposed simple explanations based on isolated molecular properties.[155] The enormous number of different empirical potentials that have been developed for water[156] is a convincing testament both to the complexity of the problem and the enduring interest it has attracted.

Jones *et al.*[147] introduced the first model for the water molecule, employing an embedded QDO to describe the electronic responses. It consisted of a rigid molecular skeleton comprising four charged sites, a single QDO tethered to the frame, and a short-ranged repulsive pair potential, as depicted in Fig. 8. It was parameterized to a set of gas phase monomer and dimer properties, with no further condensed-phase information included.

Sokhan *et al.*[157] conducted the first survey of this model's phase behavior over a wide range of thermodynamic conditions using the path integral methods described in Sec. V F. It was shown that

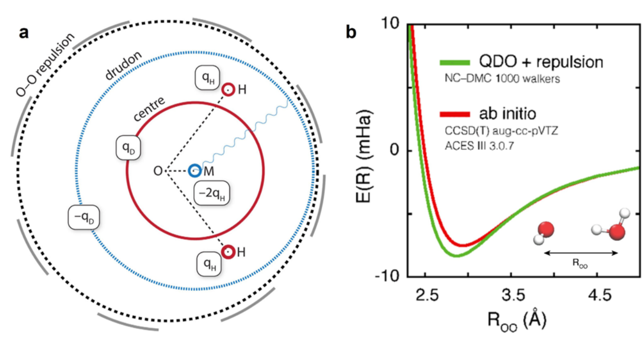


**FIG. 8.** (a) The QDO model for water as developed by Jones *et al.*[147] and applied by Sokhan *et al.* for liquid, vapor, and ice phases,[157] among others, and further explored into the supercritical regime.[158] The QDO is centered on the site labeled as "M" in the left panel. (b) Isotropic, pairwise repulsion is obtained by fitting the dimer energy calculated by *ab initio* quantum chemistry methods. Adapted with permission from Cipcigan *et al.*, Rev. Mod. Phys. **91**, 025003 (2019), including further references. Copyright 2019 American Physical Society.

the electronic responses of the model were sufficient for accurate phase equilibria, dielectric constant along the liquid–vapor coexistence curve, critical point, structure of a proton-ordered ice, and the temperature of maximum density, all to arise naturally as emergent properties of the model. The responses of the molecular dipole and quadrupole moments were also monitored at all state points, revealing significant variations. This suggests that the transferability failure modes inherent in other water models likely arise from the neglect of the electronic responses of the sort that the QDO construction remedies.

Subsequent studies explored the model's performance in the supercritical,[158] supercooled,[159] and interfacial[160] regimes, further establishing the QDO as a rare example of extreme transferability in a conceptually intuitive and computationally efficient coarse-grained model.

### C. QDO-based modeling of vdW dispersion

QDO-based models provide an ideal platform[7,126] for the accurate and efficient description of many-body vdW effects. Many-body features are highly system-dependent and hardly predictable *a priori* via semiclassical force fields. For example, even the $C_6$ dispersion coefficients of noble gases can change substantially due to the presence of neighboring atoms.[120,127] More dramatic changes in dispersion coefficients are seen due to nearby charges, the formation of hydrogen bonds, and differences in chemical bonding environments.[120,127,161] These quantitative and qualitative deviations from pairwise predictions affect multiple physical properties, such as energetics, dynamics, and response.

In bulk crystals and quasi-isotropic structures, screening effects often lead to a net reduction in vdW forces.[129] For example, in Si clusters of increasing size, collective polarization–depolarization effects effectively reduce[7] the $C_6$ Hamaker constants[162] by over 20%. A substantial reduction of $C_6$ was also observed when compressing layered materials[127,161] and for atoms within the interior of a bulk metal, compared to its surface.[163] Sizable screening (with more than 10% reduction in the vdW binding energy) was also revealed[164] in large supramolecular systems. Perturbative analysis indicates that the energy expansion up to $N$-body terms is slowly converging with respect to $N$: three-body terms alone tend to overestimate the energy reduction and are largely renormalized by higher-order corrections. As a result, the full many-body treatment of vdW interactions enabled by QDO-based modeling is also essential for the accurate description of molecular crystals.[165] Intuitively, screening can be associated with the coexistence of collective environmental polarizations propagating along different directions: on average, the absence of a specific directionality in polarizations implies cancellation effects, with reduction of the vdW coupling.

Such screening effects, along with the decrease in the vdW interaction, carry over to the case of biomolecular systems, where their complex structure and structural transitions lead to interesting ramifications. First, a general account for vdW dispersion is crucial in order to correctly describe the folding of proteins into secondary structures.[166,167] When considering a QDO-based many-body description compared to a simplified pairwise treatment, however, one finds a complex interplay of structure-dependent many-body (screening) effects. In the gas phase, a pairwise picture overstabilizes the more compact, folded state. When embedding the protein in an unstructured aqueous environment, the protein–water vdW interaction counteracts this effect, restabilizing the native state.[40]

A very different phenomenology is encountered instead in highly anisotropic systems. In sufficiently extended structures, anisotropy can support the onset of *polarization waves* with specific directionality. This consistently occurs[168–170] in 1- and 2-dimensional nanostructures (including atomic chains and 2D nanomaterials), where self-sustained polarization modes exhibit long-range wave-like patterns. Polarization waves can substantially enhance vdW couplings, producing anti-screening. At the same time, the power-law scaling of inter-fragment dispersion forces can witness major deviations from pairwise predictions, leading to a slower decay and, therefore, enhancing the effective vdW range.[168] Similarly, while generally reducing the absolute vdW interaction energy in proteins, analysis of the protein–water interaction suggests that many-body interactions lead to a considerable increase in the interaction range.[40]

In hybrid organic/inorganic systems such as metal-organic interfaces, QDO-based vdW dispersion models such as vdW$^{\mathrm{surf}}$,[171] MBD,[7,129] and MBD-NL,[93] coupled with generalized gradient approximation and hybrid functionals, provide an accurate prediction of the equilibrium structure and stability.[172,173] In several well-defined systems, including benzene and azobenzene adsorbed on an Ag(111) surface and graphene on Cu(111), MBD-based models closely match experimentally measured adsorption heights and energies,[174–176] as they reliably capture screening in the metal. Upon delamination of a graphene sheet from a Si substrate, strikingly long-ranged vdW adhesive stress with a near micrometer range was observed.[177] This effect could only be rationalized[178] when structural relaxation is governed by full many-body vdW forces. In fact, many-body contributions turn out to amplify the force susceptibility to atomic displacements.[179]

Most recently, QDOs were also coupled to interatomic charge-hopping terms to accurately describe[180] low-dimensional metallic systems. The ensuing MBD+C approach accounts for electronic charge fluctuations up to an infinite range. Charge-fluctuation waves with infinite range can further alter[181] the power-law scaling of dispersion forces up to infinite distance.

Therefore, QDO-based models play a central role in accurately capturing the complex, system-dependent nature of van der Waals dispersion, enabling a unified many-body treatment that accounts for both screening and anti-screening effects across diverse material classes. Their ability to describe collective polarization phenomena and long-range charge fluctuations makes them indispensable for modeling vdW interactions in both isotropic and highly anisotropic systems.

### D. Universal QDO-based van der Waals potentials

The QDO model's versatility in describing intermolecular interactions, as outlined in Sec. III, makes it a powerful foundation for designing fully analytical, QDO-based molecular force fields. Building on this foundation, a recent study by Khabibrakhmanov *et al.*[24] introduced a universal vdW potential based on the QDO model. This vdW-QDO potential is fully analytical, widely transferable across the periodic table, and parameterized using only minimal atomic input, making it a compelling alternative to the empirically tuned Lennard-Jones (LJ) potential.[182]

For noble-gas dimers, the vdW-QDO potential takes the form[24]

$$V_{\rm QDO}(R) = D_e U_{\rm QDO}^{\rm Ne}(x), \quad x = R/R_e, \tag{78}$$

where $U_{\rm QDO}^{\rm Ne}(x)$ is the dimensionless shape function derived from the $Ne_2$ dimer,

$$U_{\rm QDO}^{\rm Ne}(x) = \frac{A^*}{x} e^{-\frac{(\gamma^* x)^2}{2}} - \sum_{n=3}^{5} \frac{C_{2n}^*}{x^{2n}}. \tag{79}$$

The starred parameters are fixed and dimensionless, encapsulating the universal shape of the QDO-derived potential. The specificity is encoded through the equilibrium distance $R_e$ and the well depth $D_e$, both of which are inferred from fundamental atomic properties.

Specifically, the model requires only two atomic parameters: the dipole polarizability $\alpha_1$ and the $C_6$ coefficient, both widely tabulated across the periodic table. This is achieved using the quantum-mechanical scaling law for $R_e$ given by Eq. (36) and employing the optimized QDO parameterization scheme.[18,24] For noble gases, the dimer binding energy $D_e$ is obtained through a physically motivated semiempirical scaling law,[24]

$$D_e \approx \frac{C_6}{R_e^6}\left(1 - \frac{\beta - 5}{\beta(1+\beta)}\right), \quad \beta = \frac{\mu\omega}{\hbar} R_e^2. \tag{80}$$

The vdW-QDO potential yields excellent accuracy for noble-gas dimers (Fig. 9), achieving roughly twice the accuracy of the LJ potential in reproducing benchmark reference curves. It has been successfully extended to group II dimers and small organic molecules, demonstrating consistent performance across systems with diverse chemical bonding.[24] Most recently, it was incorporated into the SO3LR MLFF[36] as a physically grounded long-range correction for dispersion interactions, otherwise missing from the local neural network representations. These advancements position the vdW-QDO as a promising replacement for the LJ potential in classical force fields, especially for biomolecular and soft matter simulations where accurate treatment of vdW interactions is critical.

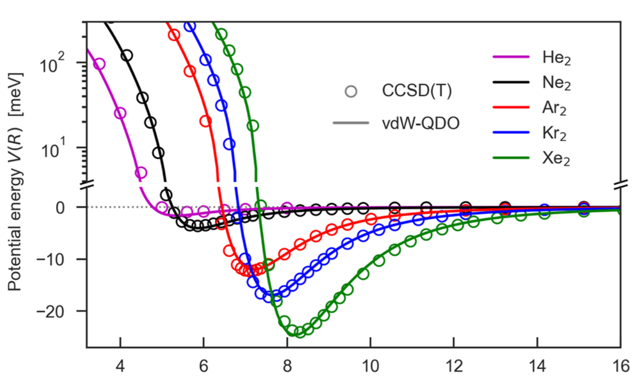


**FIG. 9.** vdW-QDO potential (solid lines) for noble-gas dimers as compared to the reference CCSD(T) calculations (circles). Reproduced with permission from A. Khabibrakhmanov, D. V. Fedorov, and A. Tkatchenko, J. Chem. Theory Comput. **19**, 7895 (2023). Copyright 2023 American Chemical Society.

## VII. EXTENSIONS TO THE QDO MODEL

The QDO model offers substantial advantages for the analysis of noncovalent interactions due to its analytic form, computational efficiency, and extendability to many-body interactions. It captures essential features of molecular and material polarization responses while incorporating both exchange-repulsion and dispersion forces. Owing to its simplicity and physical transparency—with only three parameters—the QDO model has proven to be both powerful and broadly applicable. Nonetheless, as with any physical model, the QDO framework has its limitations, and opportunities remain to refine its accuracy and to broaden its applicability.

In this section, we briefly discuss key limitations and promising directions for enhancement. Section VII A addresses the role and potential impact of nonlinear effects that are neglected in the QDO model. Section VII B reviews recent developments involving anisotropic QDOs. Finally, Sec. VII C outlines further refinements aimed at extending the model's scope and accuracy.

### A. Nonlinear response effects and the QDO model

As discussed in Sec. III H, the QDO model provides only a partial representation of nonlinear molecular responses. For example, an isotropic quantum Drude oscillator has a non-zero **B** tensor (dipole–dipole–quadrupole hyperpolarizability),[59] enabling it to capture the change in the vdW interaction energy that is linear in an external electric field for an interacting pair of different oscillators.[73,74]

However, the QDO model lacks a $\gamma$ hyperpolarizability tensor, which characterizes the quartic response to a field.[4] This can be seen from the energy shift of a single QDO in an external field given by Eq. (27), which is strictly quadratic, meaning that $\gamma$ is zero within the QDO model. Somewhat paradoxically, when an applied field acts on a pair of atoms or molecules, the $\gamma$ hyperpolarizability appears in a term in the pair dispersion energy that is quadratic in the applied field.[26,74] This contribution to the pair energy incorporates the response of each atom or molecule to the spontaneously fluctuating charge density of its interaction partner, as well as the response to the applied field. As a result, the net effect is quartic in the total local field but only quadratic in the applied field. Therefore, the QDO model does not capture all of the effects on the interaction energy that are quadratic in an external field. Here, we briefly assess the relative importance of $\gamma$ for intermolecular forces in the presence of applied fields.

In the classical dipole-induced dipole (DID) model, the first-order change in polarizability $\Delta\alpha$ scales as $R^{-3}$ with the intermolecular distance $R$, though the trace of $\Delta\alpha$ vanishes at first order.[183] At second order, classically, the isotropic average of $\Delta\alpha$ varies as $R^{-6}$. The dispersion-induced changes in polarizability, which depend on $\gamma$, have the same $R^{-6}$ dependence, suggesting that they might contribute significantly to the isotropically averaged $\Delta\alpha$ due to interactions. Two mechanisms cause this change: (1) each molecule is hyperpolarized by the concerted action of the applied field and the fluctuating field from the neighboring molecule, and (2) correlations in the charge densities of each molecule are altered by the applied field at second order, which changes the dispersion energy in the field.[73,74]

**TABLE I.** Estimated dipole-induced-dipole contributions $\alpha_c$ and dispersion contributions $\alpha_d$ to the terms that vary as $R^{-6}$ in the isotropic average of the pair polarizability for interacting noble gas atoms. Values are listed in atomic units.

| Species | He | Ne | Ar | Kr | Xe |
|---|---|---|---|---|---|
| $\alpha$ | 1.383[a] | 2.697[b] | 11.22[b] | 16.80[b] | 27.06[b] |
| $\alpha_{\text{exp}}$ | 1.40[c] | 2.669[d] | 11.08[d] | 16.79[d] | 27.16[d] |
| $\gamma$ | 43.6[e] | 115.1[b] | 1277[b] | 2756[b] | 7037[b] |
| $\gamma_{\text{exp}}$ | 44.1[f] | 119[g] | 1167[g] | 2600[g] | 6888[g] |
| $C_6$ | 1.4612[h] | 6.3826[h] | 64.30[h] | 129.6[h] | 285.9[h] |
| $\alpha_c$ | 10.58 | 78.47 | 5650 | 18 970 | 79 260 |
| $\alpha_d$ | 25.6 | 151.3 | 4066 | 11 810 | 41 300 |
| $\alpha_{\text{tot}}$ | 36.2 | 229.8 | 9716 | 30 780 | 120 600 |
| $\alpha_d/\alpha_{\text{tot}}$ (%) | 70.7 | 65.84 | 41.85 | 38.37 | 34.25 |
| $\gamma/\alpha^2$ | 22.80 | 15.82 | 10.14 | 9.765 | 9.610 |

[a]Reference 186.
[b]Reference 184.
[c]Reference 187.
[d]Reference 188.
[e]Reference 189.
[f]Reference 190.
[g]Reference 191.
[h]Reference 113.

Hunt and Bohr[73,74] developed the theory of vdW dispersion effects on polarizabilities, providing an accurate estimate of the isotropic average pair polarizability for atoms $A$ and $B$ as:[73]

$$\alpha_{AB} = \alpha_A + \alpha_B + \left[2\alpha_A^2\alpha_B + 2\alpha_B^2\alpha_A \;+ (5/18)C_6^{AB}\left(\gamma^A/\alpha^A + \gamma^B/\alpha^B\right)\right]/R^6. \tag{81}$$

The first two terms within the brackets describe the classical electrostatic contribution[183] $\alpha_c R^{-6} = \left[2\alpha_A^2\alpha_B + 2\alpha_B^2\alpha_A\right]R^{-6}$, while the last term accounts for the dispersion contribution $\alpha_d R^{-6} = \alpha_{AB} - \alpha_A - \alpha_B - \alpha_c R^{-6}$. The QDO model captures only $\alpha_c$. Table I presents QDO coefficients $\alpha_c$ and the dispersion coefficients $\alpha_d$ for the polarizability of noble-gas dimers (He to Xe) calculated using Eq. (81) along with the total interaction-induced change $\alpha_{\text{tot}}R^{-6}$. For helium and neon, $\alpha_d$ is about twice the QDO coefficient $\alpha_c$, with this ratio diminishing for the heavier noble gases yet remaining at ~50% for xenon. Likewise for radon, based on the values of $\alpha$ and $\gamma$ from Nakajima and Hirao[184] and the dispersion energy coefficient $C_6$ from Sulzer, Norman, and Saue,[185] the dispersion contribution to the isotropically averaged polarizability $\alpha_d R^{-6}$ is still ~50% of the QDO contribution $\alpha_c R^{-6}$.

An alternative measure, the ratio $\gamma/\alpha^2$, shows a linear correlation with $\alpha_d/\alpha_{\text{tot}}$, both decreasing with the atomic polarizability. The ratio of the isotropically averaged value of $\gamma$ to the square of the isotropically averaged polarizability $\alpha$ is ~6.5 for the nitrogen[192] and oxygen[193] molecules. The ratio is close to five for the bromine molecule[194] and close to four for the chlorine molecule,[195] dichloroethyne, dibromoethyne, and diiodoethyne.[196] The ratio drops below one for benzene,[197] and it is also estimated to drop below one for $C_{60}$ fullerene.[198] These findings suggest that hyperpolarization effects have less impact on larger, more polarizable molecules, so omitting them in the QDO model likely produces only small inaccuracies in the interactions of larger molecules in applied fields. Moreover, if needed, hyperpolarizabilities could be assigned to the oscillators in the QDO model.

## B. Anisotropic QDOs

Practically all QDO-based models discussed in Secs. II–VI rely on isotropic QDOs to represent atoms, atoms-in-molecules, or even small symmetric molecules. For such cases, a single characteristic frequency sufficiently captures the behavior of the oscillator. However, when a higher degree of coarse-graining is desired—such as when representing complex molecular fragments with a single oscillator—an anisotropic version of the QDO model becomes necessary to accurately capture the directional dependence of the fragment's response.

In a recent development, anisotropic QDOs have been employed to model the vdW interaction between molecular fragments.[199,200] The key modification is the generalization of the QDO potential from isotropic to anisotropic form, characterized by three distinct harmonic frequencies $\{\omega_x, \omega_y, \omega_z\}$ along the Cartesian axes. This anisotropic potential allows the QDO to reflect the inherent asymmetry of a fragment's electronic distribution and polarizability.

The QDOs are coupled through a dipole–dipole interaction tensor, which is modified by an anisotropic damping function. The anisotropic vdW radii required for damping are derived using a scaling relation based on the molecular polarizability tensor, extending the isotropic QDO scaling law of Eq. (35) to an anisotropic context.[199] Dispersion interaction energies are then obtained via second-order PT, offering an accurate yet computationally efficient approach to reproduce the intermolecular interaction energies of a wide variety of molecular dimers.

Importantly, these studies[199,200] also establish practical guidelines regarding the maximum fragment size—both in terms of geometric extent and electron count—that can be effectively modeled using a single anisotropic QDO. Further improvements to this approach could involve incorporating higher-order multipole corrections to the interaction potential, as well as developing on-the-fly QDO parameterization schemes.[199]

Overall, anisotropic QDOs offer a promising route for designing coarse-grained, physically grounded interaction potentials, making them particularly well-suited for use in quantum–classical molecular dynamics simulations.

## C. Further potential improvements to the QDO model

One fundamental limitation lies in the restricted parameter set of the QDO model, which inherently constrains accuracy for properties beyond those directly parameterized. For example, when parameterized using only $\alpha_1$ and $C_6$, the model predicts higher-order polarizabilities $\alpha_2$ and $\alpha_3$ of Eq. (7) and dispersion coefficients $C_8$ and $C_{10}$ of Eq. (9) with errors typically in the range of 20%–30%.[18] Although increasing the number of parameters could, in principle, improve the accuracy of these higher-order properties, doing so risks undermining the model's transferability and physical transparency.

A promising approach to mitigate this trade-off is to incorporate the (effective) number of valence electrons, $N_{\text{val}}$, as an additional physical input. Both atomic polarizabilities and dispersion coefficients exhibit clear trends with $N_{\text{val}}$,[81,82] suggesting that a systematic renormalization of the Thomas–Reiche–Kuhn sum rule for oscillator strengths[81] could offer a path to enhanced model flexibility. Alternatively, introducing an "electrostatic" charge $q_{\mathrm{elst}}$ in parallel with the standard QDO "response" charge $q$ could enhance electrostatic modeling without compromising the treatment of polarization and dispersion.

Another promising direction is refining QDO models with quadrupole and higher-order multipole couplings. Existing extensions[30–32] have introduced empirical quadrupole interactions to capture beyond-dipole features of the polarization response. However, despite these efforts, quadrupole-coupled QDO models have not yet demonstrated systematic improvements over dipole-only variants, particularly in the context of predictive many-body calculations. Future studies may benefit from deriving higher-order couplings from first principles, ensuring consistency with the underlying quantum mechanical symmetries and constraints. Such developments could enable the QDO model to accurately describe finer aspects of intermolecular forces, particularly in strongly anisotropic systems and under external fields.

In short, while the current QDO model already provides a remarkably successful minimal description of noncovalent interactions, further enhancements, whether through enriched parameterizations, better electrostatics, or controlled multipolar couplings, offer exciting prospects for extending its applicability to an even broader range of physical and chemical phenomena.

## VIII. SUMMARY AND OUTLOOK

The quantum Drude oscillator (QDO) model has emerged as a versatile and physically transparent framework for modeling noncovalent interactions across molecular and material systems. Its compact parameterization, analytic tractability, and intrinsic extendability to many-body effects position it as a valuable tool bridging accurate quantum-mechanical descriptions and computational efficiency. By capturing essential elements of polarization, dispersion, and exchange-repulsion within a unified picture, the QDO model provides insights not only into binding energies but also into the underlying electronic responses that govern intermolecular forces.

In this review, we have outlined the theoretical foundations of the QDO approach, detailed its practical parameterizations, and highlighted its successful applications, ranging from pairwise interactions to many-body systems and dynamical simulations. Special emphasis was placed on the physical interpretability of the model, which connects fundamental response properties—such as polarizabilities and dispersion coefficients—to simple, transferable parameters. We also discussed how the QDO model naturally incorporates many-body dispersion effects through electrodynamic coupling, offering systematic improvements over additive models without significantly increasing computational costs.

At the same time, we critically examined the current limitations of the model, particularly regarding its predictive power for higher-order response functions and dispersion coefficients. While the basic three-parameter formulation captures leading-order behaviors with already impressive and somewhat unexpected accuracy, refinements are still desirable for more demanding applications.

Looking ahead, several exciting directions can be outlined. First, the QDO model could be reverse-engineered to incorporate key concepts from density functional theory, such as orbitals, density matrices, and related quantities. This could open the door to novel and yet unexplored applications of the QDO model. Second, the Coulomb-coupled QDOs represent an intriguing minimal model for a covalent bond, as recently illustrated in Refs. 28 and 201. Finally, extending the QDO framework to describe not only ground-state but also excited-state properties represents an open frontier, potentially broadening its applicability to optical properties,[136] excitonic coupling,[136] charge-transfer processes, and beyond.

Importantly, with this review, we hope to lay a cornerstone for the growing QDO community, which has emerged from the intensive efforts of numerous research groups over the past decades. We have endeavored to collect and highlight the most significant references and research directions related to the QDO model. Nevertheless, given the vast body of work accomplished in this field, it is almost inevitable that some important contributions may have been inadvertently overlooked.

In conclusion, the QDO model stands as a powerful yet flexible platform for the study of noncovalent interactions. Its continued development and integration into multiscale modeling frameworks will undoubtedly deepen our understanding of weak forces in chemistry, biology, and materials science and inspire new approaches for controlling and engineering intermolecular interactions across different disciplines.

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## AUTHOR DECLARATIONS

### Conflict of Interest

The authors have no conflicts to disclose.

### Author Contributions

**Almaz Khabibrakhmanov**: Conceptualization (equal); Project administration (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (lead). **Dmitry V. Fedorov**:

Conceptualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Alberto Ambrosetti**: Writing – review & editing (equal). **Jason Crain**: Writing – review & editing (equal). **Katharine L. C. Hunt**: Writing – review & editing (equal). **Erin R. Johnson**: Writing – review & editing (equal). **Kenneth D. Jordan**: Writing – review & editing (equal). **Szabolcs Góger**: Writing – review & editing (equal). **Matteo Gori**: Writing – review & editing (equal). **Mohammad Reza Karimpour**: Writing – review & editing (equal). **Reinhard J. Maurer**: Writing – review & editing (equal). **Mainak Sadhukhan**: Writing – review & editing (equal). **Martin Stöhr**: Writing – review & editing (equal). **Alexandre Tkatchenko**: Conceptualization (lead); Project administration (equal); Writing – original draft (equal); Writing – review & editing (equal).

## DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

## APPENDIX: EXPECTATION VALUES IN THE MBD MODEL

As pointed out in Sec. V D, the MBD model not only provides (interaction) energies but also allows access to a more in-depth analysis by studying the corresponding wave function of the dipole-coupled QDOs. One key aspect in this regard is the evaluation of expectation values. On the basis of the $3N$ collective oscillation modes, $\boldsymbol{\zeta} = \mathbf{C}\,\boldsymbol{\xi}$, where $\mathbf{C}$ is the matrix for the normal mode transformation, the MBD Hamiltonian (63) is a set of uncoupled 1D oscillators. Hence, we can easily find the corresponding wave function as

$$\begin{aligned}\Psi_{\mathrm{MBD}} &= \prod_{k=1}^{3N}\left(\frac{\tilde{\omega}_k}{\pi}\right)^{1/4}\exp\left(-\frac{\tilde{\omega}_k}{2}\|\zeta_k\|^2\right)\\ &= \left(\prod_{k=1}^{3N}\left(\frac{\tilde{\omega}_k}{\pi}\right)^{1/4}\right)\exp\left(-\frac{1}{2}\sum_{j,l}\xi_j\,\Omega_{jl}\,\xi_l\right),\end{aligned} \tag{A1}$$

where we introduced $\Omega_{jl} = \sum_k \tilde{\omega}_k \mathbf{C}_{jk}\mathbf{C}_{lk}$ in order to represent $\Psi$ in the basis of the original mass-weighted displacements, $\boldsymbol{\xi}$.

We then consider the expectation value of a given operator $\hat{O}_{\mathcal{X}} \equiv \hat{O}(\{\mathbf{r}_X\})$, i.e., an operator acting on all Drude particles in the set $\mathcal{X}$. In the context of this review, we limit our description to operators that are functions of the position operator, such that the expectation value simplifies to

$$\begin{aligned}O_{\mathcal{X}} &= \langle\Psi_{\mathrm{MBD}}|\hat{O}_{\mathcal{X}}|\Psi_{\mathrm{MBD}}\rangle\\ &= \left(\prod_{k=1}^{3N}\sqrt{\frac{\tilde{\omega}_k}{\pi}}\right)\int\hat{O}_{\mathcal{X}}\,\exp\left(-\overbrace{\sum_{j,l}\xi_j\,\Omega_{jl}\,\xi_l}^{\xi^{\mathrm{T}}\Omega\,\xi}\right)d\boldsymbol{\xi}.\end{aligned} \tag{A2}$$

For example, this includes the single-particle charge density operator, $\hat{n}(\mathbf{r}_A) = q_A\delta(\mathbf{r}-\mathbf{r}_A)$, where $\mathcal{X} = \{A\}$, or the two-particle Coulomb operator, $\hat{V}(\mathbf{r}_A,\mathbf{r}_B) = q_Aq_B/\|\mathbf{r}_A-\mathbf{r}_B\|$, where $\mathcal{X} = \{A,B\}$. We then split the exponent into contributions that $\hat{O}_{\mathcal{X}}$ acts on and that it does not, meaning into mass-weighted displacements of Drudons in the set $\mathcal{X}$ and those not in $\mathcal{X}$,

$$\begin{aligned}\boldsymbol{\xi}^{\mathrm{T}}\boldsymbol{\Omega}\,\boldsymbol{\xi} &= \sum_{\substack{j\notin\mathcal{X}\\ l\notin\mathcal{X}}}\Omega_{jl}\xi_j\xi_l + 2\sum_{\substack{p\in\mathcal{X}\\ q\notin\mathcal{X}}}\Omega_{pq}\xi_p\xi_q + \sum_{\substack{r\in\mathcal{X}\\ s\in\mathcal{X}}}\Omega_{rs}\zeta_r\zeta_s\\ &= \boldsymbol{\xi}_{\bar{\mathcal{X}}}^{\mathrm{T}}\,\boldsymbol{\Omega}_{\bar{\mathcal{X}}\bar{\mathcal{X}}}\,\boldsymbol{\xi}_{\bar{\mathcal{X}}} + 2\,\boldsymbol{\xi}_{\bar{\mathcal{X}}}^{\mathrm{T}}\,\boldsymbol{\Omega}_{\bar{\mathcal{X}}\mathcal{X}}\,\boldsymbol{\xi}_{\mathcal{X}} + \boldsymbol{\xi}_{\mathcal{X}}\,\boldsymbol{\Omega}_{\mathcal{X}\mathcal{X}}\,\boldsymbol{\xi}_{\mathcal{X}}.\end{aligned} \tag{A3}$$

Here, $\boldsymbol{\xi}_{\mathcal{X}}$ ("$\xi$ $\mathcal{X}$") and $\boldsymbol{\xi}_{\bar{\mathcal{X}}}$ ("$\xi$ not-$\mathcal{X}$") are the subvectors of the $3N$-dimensional displacement vector $\boldsymbol{\xi}$ that correspond to Drudons in $\mathcal{X}$ and not in $\mathcal{X}$, respectively. Analogously, $\boldsymbol{\Omega}_{\mathcal{X}\mathcal{X}}$ describes the submatrix of $\boldsymbol{\Omega}$, where all rows and columns refer to Drudons in $\mathcal{X}$, and $\boldsymbol{\Omega}_{\bar{\mathcal{X}}\bar{\mathcal{X}}}$ is the submatrix where none of the rows or columns refer to Drudons in $\mathcal{X}$. Finally, $\boldsymbol{\Omega}_{\bar{\mathcal{X}}\mathcal{X}}$ is the off-diagonal submatrix where each row refers to Drudons not in $\mathcal{X}$ and each column to Drudons that are in $\mathcal{X}$. For a simple illustration, consider the case where $\mathcal{X}$ denotes the set of the first two Drude oscillators. In this case, the full matrix $\boldsymbol{\Omega}$ is partitioned as follows:

$$\boldsymbol{\Omega} = \left(\begin{array}{c|c}\boldsymbol{\Omega}_{\mathcal{X}\mathcal{X}} & \boldsymbol{\Omega}_{\mathcal{X}\bar{\mathcal{X}}}\\ \hline \boldsymbol{\Omega}_{\bar{\mathcal{X}}\mathcal{X}} & \boldsymbol{\Omega}_{\bar{\mathcal{X}}\bar{\mathcal{X}}}\end{array}\right) \in \mathbb{R}^{3N\times 3N}, \tag{A4}$$

where, in this example, the submatrix $\boldsymbol{\Omega}_{\mathcal{X}\mathcal{X}}$ would be $(6\times 6)$-dimensional and the rest accordingly.

Next, we note that the final expression in Eq. (A3) follows the form $\mathbf{x}^{\mathrm{T}}\mathbf{M}\,\mathbf{x} + \mathbf{x}^{\mathrm{T}}\mathbf{v} + c = (\mathbf{x} + \mathbf{M}^{-1}\mathbf{v}/2)^{\mathrm{T}}\mathbf{M}(\mathbf{x} + \mathbf{M}^{-1}\mathbf{v}/2) + c - \mathbf{v}^{\mathrm{T}}\mathbf{M}^{-1}\mathbf{v}/4$. Completing the square in the same form for Eq. (A3), we arrive at

$$\boldsymbol{\xi}^{\mathrm{T}}\boldsymbol{\Omega}\,\boldsymbol{\xi} = \boldsymbol{\gamma}_{\mathcal{X}}^{\mathrm{T}}\,\boldsymbol{\Omega}_{\bar{\mathcal{X}}\bar{\mathcal{X}}}\,\boldsymbol{\gamma}_{\mathcal{X}} + \boldsymbol{\xi}_{\mathcal{X}}^{\mathrm{T}}\boldsymbol{\Gamma}_{\mathcal{X}}\boldsymbol{\xi}_{\mathcal{X}}, \tag{A5}$$

where we have introduced $\boldsymbol{\gamma}_{\mathcal{X}} = \boldsymbol{\xi}_{\bar{\mathcal{X}}} + \boldsymbol{\Omega}_{\bar{\mathcal{X}}\bar{\mathcal{X}}}^{-1}\,\boldsymbol{\Omega}_{\bar{\mathcal{X}}\mathcal{X}}\,\boldsymbol{\xi}_{\mathcal{X}}$ and the Schur complement $\boldsymbol{\Gamma}_{\mathcal{X}} = \boldsymbol{\Omega}_{\mathcal{X}\mathcal{X}} - \boldsymbol{\Omega}_{\mathcal{X}\bar{\mathcal{X}}}\,\boldsymbol{\Omega}_{\bar{\mathcal{X}}\bar{\mathcal{X}}}^{-1}\,\boldsymbol{\Omega}_{\bar{\mathcal{X}}\mathcal{X}}$. This finally allows us to factorize the exponent in Eq. (A2) and obtain (note that we used $d\boldsymbol{\xi}_{\bar{\mathcal{X}}} = d\boldsymbol{\gamma}_{\mathcal{X}}$)

$$\begin{aligned}O_{\mathcal{X}} &= \left(\prod_{k=1}^{3N}\sqrt{\frac{\tilde{\omega}_k}{\pi}}\right)\int\hat{O}_{\mathcal{X}}\,\exp\left(-\boldsymbol{\xi}_{\mathcal{X}}^{\mathrm{T}}\boldsymbol{\Gamma}_{\mathcal{X}}\boldsymbol{\xi}_{\mathcal{X}}\right)\\ &\quad\times\left(\int e^{-\boldsymbol{\gamma}_{\mathcal{X}}^{\mathrm{T}}\,\boldsymbol{\Omega}_{\bar{\mathcal{X}}\bar{\mathcal{X}}}\,\boldsymbol{\gamma}_{\mathcal{X}}}\,d\boldsymbol{\gamma}_{\mathcal{X}}\right)d\boldsymbol{\xi}_{\mathcal{X}}.\end{aligned} \tag{A6}$$

The integral in parentheses in the second line above is a standard integral over an anisotropic Gaussian and can be solved analytically. This yields the final expression

$$O_{\mathcal{X}} = \left(\prod_{k=1}^{3N}\sqrt{\tilde{\omega}_k}\right)\sqrt{\frac{\pi^{-3|\mathcal{X}|}}{\det\{\boldsymbol{\Omega}_{\bar{\mathcal{X}}\bar{\mathcal{X}}}\}}}\int\hat{O}_{\mathcal{X}}\,\exp\left(-\boldsymbol{\xi}_{\mathcal{X}}^{\mathrm{T}}\boldsymbol{\Gamma}_{\mathcal{X}}\boldsymbol{\xi}_{\mathcal{X}}\right)d\boldsymbol{\xi}_{\mathcal{X}}, \tag{A7}$$

where $|\mathcal{X}|$ is the cardinality of $\mathcal{X}$, or in other words, the number of Drude particles that the operator $\hat{O}_{\mathcal{X}}$ acts on. Finally, we note that $\prod_k\sqrt{\tilde{\omega}_k} = \sqrt{\det\{\boldsymbol{\Omega}\}}$ and that $\det\{\boldsymbol{\Omega}\}/\det\{\boldsymbol{\Omega}_{\bar{\mathcal{X}}\bar{\mathcal{X}}}\} = \det\{\boldsymbol{\Omega}_{\mathcal{X}\mathcal{X}} - \boldsymbol{\Omega}_{\mathcal{X}\bar{\mathcal{X}}}\,\boldsymbol{\Omega}_{\bar{\mathcal{X}}\bar{\mathcal{X}}}^{-1}\,\boldsymbol{\Omega}_{\bar{\mathcal{X}}\mathcal{X}}\} = \det\{\boldsymbol{\Gamma}_{\mathcal{X}}\}$. For the latter, note that $\boldsymbol{\Omega}_{\mathcal{X}\mathcal{X}}$, $\boldsymbol{\Omega}_{\bar{\mathcal{X}}\mathcal{X}}$, and $\boldsymbol{\Omega}_{\bar{\mathcal{X}}\bar{\mathcal{X}}}$ are the block-wise components of the full matrix $\boldsymbol{\Omega}$. Block-LDU decomposition of $\boldsymbol{\Omega}$ and subsequent evaluation of the determinant show the required equality. With this, we can simplify Eq. (A7) to

$$O_{\mathcal{X}} = \sqrt{\frac{\det\{\boldsymbol{\Gamma}_{\mathcal{X}}\}}{\pi^{3|\mathcal{X}|}}}\int\hat{O}_{\mathcal{X}}\,\exp\left(-\boldsymbol{\xi}_{\mathcal{X}}^{\mathrm{T}}\boldsymbol{\Gamma}_{\mathcal{X}}\boldsymbol{\xi}_{\mathcal{X}}\right)d\boldsymbol{\xi}_{\mathcal{X}}. \tag{A8}$$

Therefore, all that remains to obtain the final expectation value is to evaluate the above integral, including the operator's action on an anisotropic Gaussian.

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